Cosmic Quarry Mining Corporation

Insurance Feasibility Report

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GALAXY GENERAL INSURANCE

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1. Executive Summary

1.1 Links to Project Objectives

The provided recommendations reflect the project objectives through the way in which they were informed by statistical modelling results, with statistical summary tables provided in *Appendix A*. Each hazard-coverage product has been tiered to reflect the differing associated risk levels, but also the need to manage extreme tail events for capital protection. Annual premium structures for each product have been designed to cover expected losses and risk margins whilst simultaneously stabilising long-term profitability in cases of volatility. Additionally, by utilising the Q-RISK Engine, Galaxy General Insurance (GGI) is able to boost efficiency.

The recommended three-tier Business Interruption (BI) product captures high-frequency moderate losses (Core/Enhanced packages) but also extreme tail risk (Catastrophic package), with pricing that incorporates station-level and solar system-level risk profiles. By incorporating the BI products, GGI's portfolio will be instantly enhanced, adding a high premium, high-margin product line that diversifies across risk tiers to cover against BI. GGI is strongly recommended to implement the proposed Equipment Failure (EF) products into their portfolio given the cruciality of machinery for mining output. The offered EF product follows a three-tiered structure which corresponds with variation across machine-level risks. It varies by environment and machine type, and controls adverse selection through deductibles and waiting periods. GGI can improve their portfolio diversification in a market of high demand, leading to a stable revenue stream as claims are frequent yet manageable.

GGI is also recommended to incorporate our proposed Workers' Compensation (WC) product into their insurance portfolio, providing a long-duration and predictable liability stream, based upon pricing that reflects roles, environment, and claim duration assumptions. By taking on WC as a coverage line, GGI will further diversify their portfolio, allowing for a competitive edge in the interstellar GI market. Finally, Cargo Loss (CL) products have been recommended, providing a strong competitive advantage as the pricing is based directly upon the riskiness of the cargo route. This product line covers a wide range of extreme hazards and offers fast, lump-sum payouts. The design is especially flexible and scalable across all solar systems, allowing for GGI to easily expand.

Each product has been explicitly tailored through risk segmentation, tiered products, and parameterised via the Q-RISK Engine. The consistent tiered architectures allow for solar systems to select coverage based upon their unique hazard intensities, with premiums scaled based upon both environmental and operational risks. Additionally, the coverage triggers and exclusions have been tailored to system specific threats to minimise moral hazard and underinsurance. The Q-RISK engine supports scalability through its dynamic nature for real-time modelling and capital allocations. Additionally, scalability will be seamless as the Q-RISK Engine acts as a single platform to manage each hazard area coverage, providing the foundation for long-term portfolio growth grounded in robust modelling.

Ultimately, the recommended selection of products has been designed to expand with CQMC's growing operations, holding the potential to respond to new and evolving hazards. In following these recommendations, GGI can provide resilient, and profitable coverage across all solar systems.

1.2 Product ERM Integration to Assist Marketing & Sales Teams

Cosmic Quarry Mining Corporation (CQMC) should define their risk appetite by hazard area, recognising the differences between each solar system and their dependencies. CQMC will likely adopt a risk-based purchasing strategy that segments risk based upon hazard and solar system. At GGI, marketing and sales teams will leverage off this, offering flexible contracts that reflect

frequency vs severity, but also system-specific risk profiles. GGI will provide products specifically to integrate with CQMC's existing ERM framework, aligned specifically to the differing risk segments of each hazard coverage.

CQMC may be influenced by the potential to be able to bundle risks and negotiate portfolio-level pricing across solar systems. GGI marketing and sales teams can utilise this potential by providing flexible support in offering portfolio-level pricing, allowing CQMC to buy smarter. CQMC will be influenced by the potential to use GGI data to drive their own decision making by feeding their ERM system with insights. This will be done via claim dashboards by hazard and solar system and introduce early warning indicators. This will allow CQMC to identify emerging risks, optimise resource allocation, and adjust operations simultaneously. For GGI, this translates to strong underwriting precision, robust risk pricing, and proactive capital management. Claim patterns can be anticipated to simultaneously maintain profitability and support CQMC's operational resilience.

GGI will support CQMC's ongoing ERM cycle, extending beyond the standard annual renewals. GGI will offer a continuous review process that entails regular updates for exposure, and dynamic repricing where needed. This will allow the insurance to evolve with their risk profile, reducing the risk of becoming outdated in the market. Overall, GGI will stand out against competitors as they have the capabilities to pitch their products as integrated components of CQMC's ERM framework, rather than just standalone cover.

2. Product Design

For the four hazards, we have determined that each product will provide coverage for one year, until it is repurchased again for the following year. At the beginning of each year, one premium payment will be made, providing GGI with the needed liquidity given the nature of claims under each hazard coverage. Additionally, this structure is favourable as it simplifies the interest assumptions product requirements. Further, in utilising advanced processing technologies, GGI will provide an instantaneous lump sum benefit for every claim, as soon as it is validated by our system.

2.1 Business Interruption

BI coverage offers a three-tiered product, segmented by risk, with premiums differing per policy due to the advanced nature of the Q-RISK engine. The three tiers are Core BI Protect, Enhanced BI Plus, Catastrophic BI Ultimate. The product is tiered according to the experience of crews, load volumes and station risk. The premiums and coverage rise with each tier of the product, appearing in *Appendix B.1*. Tier-specific benefit structure details are provided in *Appendix B.1.1*.

BI Protect will only cover direct losses from full shutdowns for local stations, includes a 48-hour waiting period, and excludes high risk stations. Enhanced BI Plus adds on cover for partial interruptions, system inefficiencies, and interlinked station issues, includes a 24-hour waiting period, and does not cover multi-system failures. Catastrophic BI Ultimate uniquely covers full system-wide failures, their cascading breakdown effects, and extended periods of shutdown, exhibiting no waiting period. Tier-specific trigger details are provided in *Appendix B.1.2*.

Applying to all BI coverage tiers, we will not cover losses as a result of gradual deterioration, lack of maintenance, and defects known prior to policy inception. Negligence, operator errors, and failure to meet minimum safety standards/regulations will also be excluded. Coverage does not extend to voluntary shutdowns, planned maintenance shutdowns, or cost-cutting decisions that impact operations. If losses are unable to be verified and records are kept poorly, the coverage will also not be activated. By having strict BI exclusions, we remove high frequency and low severity

noise whilst simultaneously preventing moral hazard and the inflation of claims. Given the interconnected nature of stations within the same solar system, traditional interdependence assumptions deteriorate as single BI events can result in multiple claims and huge aggregate losses.

2.2 Equipment Failure

We offer CQMC an annual Equipment Failure program with three forms: Equipment Failure Essentials, Equipment Failure Core, and Equipment Failure Plus. Our cover includes repair or replacement costs for sudden accidental breakdown, emergency response costs, and reinstatement expenses. We tier the product by machinery risk. Deductibles, coinsurance, and waiting periods increase with risk, so lower-risk machinery receives lighter terms while higher-risk machinery such as Quantum Bores, Graviton Extractors, and upper-tier Flux Riders / FluxStream Carriers is written on stronger terms and higher annual charged premiums. Detailed machine-level schedules are provided in *Appendix C.1*.

Covered events include internal mechanical or electrical malfunction, control-system failure, debris or collision impact, radiation-linked malfunction, flare disruption, and thermal shock. We apply this trigger across all three solar systems. We exclude wear and tear, gradual deterioration, deferred maintenance, known defects existing before policy inception, and intentional misuse. Intentional damage and misuse are not covered trigger events. We recognise that the Equipment Failure fleet is not homogeneous. Risk varies materially by machine type and by solar system, so a flat product would underprice upper-tier machinery and overprice lower-tier machinery. We address this by linking product form and underwriting terms directly to modelled risk. As risk rises, we strengthen deductibles, coinsurance, and waiting periods, which makes the product both actuarially priced and operationally disciplined.

2.3 Workers' Compensation

We offer CQMC an annual Workers' Compensation product with three forms: Workers Compensation ActiveCare Essentials, Workers Compensation ActiveCare, and Workers Compensation ActiveCare Plus. The role-level premiums shown in the *Appendix C.1* represent annual premiums allocated per worker segment rather than amounts paid directly by employees. Our premium schedule also reflects more than salary alone: it incorporates role, employment type, predicted claim duration, and operating environment, which makes the pricing structure more defensible than a flat payroll-based schedule. Our cover includes wage replacement, medical treatment, and rehabilitation support. We structure wage replacement using the statutory-style basis adopted in the model, with benefits beginning at a higher replacement rate in the first entitlement period and stepping down in the second entitlement period, subject to the duration cap built into the pricing assumptions (following Australian Labour Laws). Detailed role-level schedules are provided in *Appendix D.1*.

We trigger Workers' Compensation cover when a worker suffers a work-related injury or illness during the policy period arising out of or in the course of employment with CQMC. Covered events include machinery-related injury, physical strain injury, transport-related injury, fatigue-related injury, and other occupational injury or illness arising from mining operations. We apply this trigger across all covered worker classes. Wage replacement is paid at 95% of pre-injury weekly earnings for the first 13 weeks of incapacity, then at 80% thereafter while incapacity continues, subject to the statutory weekly maximum (Đ2,662.10) and a benefit cap of 260 weeks for non-severe injuries. Cover also includes medical treatment, rehabilitation support, and permanent disability or death benefits. We exclude fraud, intentional self-harm, and circumstances outside covered employment. We recognise that the main limitation in Workers' Compensation is not the absence of system-level or worker-segment data, but the amount of extrapolation required. Bayesia System and Oryn Delta require more assumption-setting than Helionis Cluster, and workforce

counts are harder to estimate with confidence. We therefore treat workforce counts as modelled pricing exposures rather than official staffing recommendations.

2.4 Cargo Loss

For CL, a competitive pricing structure will utilise the variable 'Route Risk', to categorise shipments into 5 levels of riskiness: Low Cargo Cover, Core Transit Cover, Shipment Essentials, High Transit Cover, and Extreme Cargo Cover. CL, unlike most other hazards, does not categorise claims by solar system, thus the scalability of CL Cover to shipments in the new solar systems will be simpler. Another key component of the product is the maximum deductible of Đ31K and a limit of Đ678,000K. See *Appendix E.1* for product premiums.

The product provides coverage for external uncontrollable losses during transit. To be covered, the shipment must be accurately documented and contain no fraudulent information. Further, cargo valued below Đ50K or in exceedance Đ680,000K will not be covered, as well as if the container is older than 50 earth years. Importantly, the shipment can only travel for less than 100AU taking at maximum 60 earth months to arrive, otherwise coverage will be redacted.

3. Pricing & Capital Modelling

3.1 Business Interruption

For BI, the short-term and long-term aggregate losses are right skewed, especially in the long-term scenario, with a long tail of extreme losses to highlight the prevalent tail risk associated with BI. A detailed plot of aggregate loss distributions can be found in *Appendix B.2.1*, and coinciding important statistics can be found in *Appendix A*.

In the short-term, BI losses are strongly concentrated around the mean, sitting at roughly Đ652M and it is implied that whilst large losses exist, they exhibit less spread. When observing at the 95th and 99th percentiles, we notice that they are relatively close to the mean, sitting Đ668M and Đ675M respectively, inciting less strain upon short-term capital requirements. In the long-term, the mean aggregate loss skyrocketed to Đ3.5B and there is a wide gap to the 95th and 99th percentiles, valuing Đ3.7B to Đ3.8B respectively, implying substantial accumulation of tail risk. Furthermore the gap between the 95th and 99th percentile is larger than that of the short-term, implying that extreme events become more dangerous over time. This highlights the necessity for robust tail risk management in dealing with BI, especially as the risk profile is dominated by extreme events.

To mimic extreme scenarios, BI space stress was included to capture the effects space specific issues upon business interruptions. These scenarios have the potential to lead to extended downtime, severity spikes, frequency spikes, and correlated losses. Multi-year simulations were run, with spikes added to frequency and severity, but premium values were maintained given their inability to fluctuate with real-time risk. The BI net revenue distribution under space stress can be visualised in *Appendix B.2.2*, displaying a highly negative net revenue. This 1-in-100 worst case scenario has the propensity to wipe out the entire premium base over and over again in a single hit. This scenario is evidently tail dominated, with the distribution so far skewed that it cannot be covered without huge capital reserves or reinsurance. This is a powerful argument for the implementation of strict hazard-specific exclusions in the BI product designs.

Simulations were run with inflation rates consistently set above 5%, inflating severity, and premiums as a pricing response. When high inflation occurs consistently over a multi-year period, severity increased but the shape of the cost distribution remained similar to that of the baseline, with a slightly thicker yet still manageable tail. Additionally, the tail appears to be smooth, even in

the worst-case outcome, net revenue remains positive. This shows the strong resilience of the BI portfolio to parameter shocks.

3.2 Equipment Failure

We price Equipment Failure as the dominant earnings and capital line in the proposed portfolio. In the short term, expected annual loss is Đ247.55M, projected claims cost is Đ259.00M, charged premium is Đ385.0M, and annual net revenue is Đ65.8M. In the long term, expected annual loss remains about 247.55M, while projected claims cost rises to Đ465.07M, charged premium rises to Đ691.0M, and annual net revenue rises to Đ119.0M.

The Equipment Failure results show clear and significant deterioration across scenarios, with both expected losses and tail risk increasing from Best to Moderate to Worst_1in100 in the short and long term. Importantly, the rise in P99 and TVaR99 is more pronounced than the increase in the mean, indicating that worsening conditions do not merely shift the distribution upward, but also make extreme outcomes materially more severe. This suggests that Equipment Failure is particularly vulnerable to adverse scenarios, with risk emerging through both higher baseline losses and greater tail exposure. The effect is even stronger in the long term, where the accumulation of stress leads to substantial compounding in aggregate losses as seen in *Table 3.1*. Further details can be found in *Appendix C.2*.

Relative to the Moderate case, a maintenance backlog shock increases mean aggregate loss by 10%, a usage intensity shock by 15%, a catastrophe severity shock by 35%, and a combined operating shock by 50% in both the short term and long term. The model also allows for dependency across solar systems. More detail is provided in *Appendix C.2*.

Table 3.1 Portfolio pricing and capital summary

Hazard	Horizon	Expected annual loss (m)	Projected claims cost (m)	Charged premium (m)	Annual net revenue (m)	Moderate mean loss (m)	Moderate P99 (m)	Worst-case TVaR99 (m)
Equipment Failure	Short term	247.55	259.00	385.0	65.8	247.6	263.0	410.0
Equipment Failure	Long term	247.55	465.07	691.0	119.0	247.6	263.0	410.0

3.3 Workers' Compensation

We price Workers' Compensation as a smaller but still profitable supporting line. In the short term, expected annual loss is Đ8.03M, projected claims cost is Đ8.39M, charged premium is Đ12.5M, and annual net revenue is Đ2.13M. In the long term, projected claims cost rises to Đ14.81M, charged premium to Đ22.0M, and annual net revenue to Đ3.8M. This shows that Workers' Compensation contributes positive earnings, but on a much smaller scale than Equipment Failure.

Although Workers' Compensation produces a much tighter aggregate loss distribution than Equipment Failure, the scenario results still show clear and systematic deterioration. From Best to Worst_1in100, both the expected loss and the tail risk measures increase across the short and long term, as seen in *Table 3.2*, indicating that adverse conditions shift the entire loss distribution upward rather than simply creating a few isolated extreme outcomes. While Workers' Compensation appears less volatile, but it remains highly sensitive to worsening assumptions, particularly over the longer horizon where the cumulative effect of stress becomes more pronounced. As a result, the main risk in this line is not distributional spread alone, but the persistent escalation in overall loss levels and extreme loss exposure as conditions worsen. Further details are provided in *Appendix D.2*.

Relative to the Moderate case, a fatigue / overtime shock increases mean aggregate loss by 8%, a recovery duration shock by about 12%, a benefit severity shock by about 15%, and a combined workforce shock by about 25% in both horizons. Dependency is also retained, although its capital effect is weaker than for Equipment Failure. More detail is provided in *Appendix D.2*.

Table 3.2 Portfolio pricing and capital summary

Hazard	Horizon	Expected annual loss (m)	Projected claims cost (m)	Charged premium (m)	Annual net revenue (m)	Moderate mean loss (m)	Moderate P99 (m)	Worst-case TVaR99 (m)
Workers Compensation	Short term	8.03	8.39	12.5	2.13	8.03	9.46	13.0
Workers Compensation	Long term	8.03	14.81	22.0	3.8	8.03	9.46	13.0

3.4 Cargo Loss

A detailed plot of aggregate loss distributions can be found in *Appendix E.2.2*. The short-term losses are roughly comparable to a normal distribution, and most losses fit in a fairly narrow band, indicating fairly stable outcomes. For GGI, this gives security that the premiums are adequate and that capital reserves are more manageable under ordinary business operations. However, long-term losses follow a more dispersed multimodal distribution. This suggests that aggregate losses become more uncertain with time as inflation and interest scatter outcomes as their effects compound. It's important to understand the sensitivity of the product to economic conditions, to be able to mitigate and counter-balance these long-term risks.

Nevertheless, it is important to note that the long-term distribution is heavily limited by its assumptions that premium and severity will grow equally with inflation, as well as that interest rates will follow historical trends. Further, the simulation does not capture how transportation technology, route conditions and claim behaviours might change expected losses throughout this period. Since the CL data does not contain solar system information, the product inherently assumes dependence across solar systems such that environmental and operational shocks affect all insured shipments concurrently. The most influential variables for claim frequency and severity (see *Appendix E.2.1*) were selected as the main stress testing triggers for the CL portfolio.

Several adverse scenarios were created assess how aggregate losses are affected by key risk drivers. Environmental factors including solar radiation and debris density were seen to impact aggregate losses moderately. Operational factors including container constraints were found to have a limited impact in isolation. 'Route risk' was found to cause the biggest disturbance upon claims, driving the largest increase in loss out of all the tested stress scenarios. These results have important implications for underwriting, pricing and capital management, as they showcase product volatility under extreme events. This highlights the importance of reserving capital to act as a buffer for uncertainty. Refer to *Appendix E.2.3* for further CL stress testing details.

4. Risk Assessment

4.1 Identification of Risks by Hazard Coverage of Each Solar System

There are general risks which all four hazard areas experience regardless of the solar system. This primarily includes the risk of fraudulent claims and the risk that policyholder data may be incomplete, causing inefficiencies in pricing and increased underwriting risk.

Key BI risks include the over utilisation of CQMC's production capacity, energy failure, crew inexperience and improper maintenance, and extended shutdown periods. Cargo risks are expected to vary across solar systems, especially for debris exposure and solar flare activity. Operational risks, including route risk, vessel stress and pilot inexperience, apply regardless of the solar system. A more detailed cargo risk breakdown is included in *Appendix E.3.1*.

4.2 Business Interruption

In terms of BI, correlated risk scenarios may include system wide shutdowns, forcing many operations to stop simultaneously, supply chain disruptions, and extended recovery periods as replacement parts become limited, due to increased demand, and the interstellar distance slowing transport. This results in cascading effects upon the entire supply chain, leading to heavy claim severity as revenue is negatively affected for a prolonged time.

Ultimately, correlated risk scenarios are likely to trigger all four coverage areas simultaneously. For example, if more equipment breaks, operations decrease, revenue decreases, and business interruptions skyrocket. Correlation across hazard lines extend into coverage lines, leading to mass increases in claims as the same clients activate multiple policies at once.

For BI scenario testing, the best-case scenario entails frequency dropping by 20%, severity dropping by 30%, and returns increasing by 20%. A detailed scenario testing plot has been provided in *Appendix B.3.1*. The tight shape implies there is low volatility associate with the outcomes, and it has a mild right skew. The mean net revenue in the best-case scenario sits comfortably at Đ3.41B, showing that the BI products will thrive under such conditions. The moderate-case scenario behaves as per normal, acting as a "business-as-usual" baseline. The mean net revenue is still positive, sitting at Đ420M. This highlights the fact that the recommended BI products are not hugely profitable under baseline conditions, providing a strong case to discourage discounts. The worst-case scenario is based off of a common shock, with correlation between the risks impacting both frequency and severity simultaneously and decreasing returns due to market crashes. The mean net revenue under this case is extremely negative, lying at - Đ213B. A heavy left tail is exhibited, displaying the domination of correlated shocks upon the risk profile as opposed to less detrimental independent fluctuations.

Overall, BI scenario testing reveals that when shocks are independent, the product is stable, however it collapses when shocks become correlated.

Table 4.1 BI Top Threats

Threat	Why?	BI Impact
Space-Stress	Strongest correlation, wide failure channels	Deep systematic impact, severe operational halts
Common Shocks	Drives correlation between frequency and severity	Heavy left tail losses
Market Crashes	Removes investment buffer, doesn't directly impact frequency/severity	Net revenue collapses
Supply-chain fragility	Upstream failures spread to all dependent operations, systematic but slower moving correlation	Long duration outages with slow recovery, high severity

4.3 Equipment Failure

Risk identification differs a fair amount by solar system. In Helionis Cluster, the main threat is debris impact, micro-collision, and communications clutter, and it is also the largest Equipment Failure book, with short-term expected annual loss of about 180.0m. In Bayesia System, the main threat is radiation spike, thermal shock, and shielding failure, with short-term expected annual loss of about Đ46.4M. In Oryn Delta, the main threat is flare disruption, orbital shear, and relay outage, with short-term expected annual loss of about Đ21.6M.

Correlated risk is significant as the underlying hazards are not fully independent across solar systems. The Best case assumes smooth operations with attritional claims only, the Moderate case assumes base operating conditions, and the Worst case assumes a catastrophic correlated event across solar systems. For Equipment Failure, this moves short-term mean aggregate loss from

Đ228M in Best to Đ267M in Moderate and Đ324M in Worst, and long-term mean loss from Đ409M to Đ479M to Đ581M. Further details on these scenarios are available in *Appendix C.3*.

The top Equipment Failure threats are combined operating deterioration, catastrophe severity shock, usage intensity shock, and maintenance backlog shock. Helionis is the core scale market and the largest contributor to premium and aggregate loss. Bayesia is technically priceable, but environmental hazards justify explicit loadings. Oryn Delta is the most fragile system under downside conditions and therefore warrants the strongest underwriting discipline.

Table 4.2 EF Top threats

Hazard	Solar system	Primary threat	Short-term expected annual loss
Equipment Failure	Helionis Cluster	Debris impact / micro-collision / communications clutter	180.0m
Equipment Failure	Bayesia System	Radiation spike / thermal shock / shielding failure	46.4m
Equipment Failure	Oryn Delta	Flare disruption / orbital shear / relay outage	21.6m

4.4 Workers' Compensation

Workers' Compensation risk identification also differs by solar system. In Bayesia System, the primary threat is gravity strain, radiation exposure, and heat stress. In Helionis Cluster, the key threat is extraction injury, transport incident, and debris-linked trauma. In Oryn Delta, the main threat is low-visibility incident, fatigue, and delayed rescue. Short-term expected annual loss is highest in Helionis at about Đ3.66M, followed by Oryn Delta at about Đ2.53M, and Bayesia at about Đ1.85M.

Correlated risk is not as high as it was for Equipment Failure, but it is still relevant. The Best case assumes smooth operations with attritional claims only, the Moderate case assumes base workforce conditions, and the Worst case assumes a catastrophic correlated event. This moves short-term mean aggregate loss from Đ7.17M in Best to Đ8.39M in Moderate and Đ10.2M in Worst, and long-term mean loss from Đ12.7M to Đ14.8M to Đ18.0. More details on these scenarios are provided in *Appendix D.3.*

The top Workers' Compensation threats are combined workforce shock, benefit severity shock, recovery duration shock, and fatigue / overtime shock. Helionis remains the largest base for expected loss, but Oryn Delta remains a meaningful operational-risk system because delayed rescue, fatigue, and visibility challenges compound workforce exposure. Bayesia is smaller in absolute loss but still exposed to heat, gravity, and radiation-related strain.

Table 4.3 WC Top threats

Hazard	Solar system	Primary threat	Short-term expected annual loss
Workers' Compensation	Helionis Cluster	Extraction injury / transport incident / debris-linked trauma	3.66m
Workers' Compensation	Bayesia System	Gravity strain / radiation exposure / heat stress	1.85m
Workers' Compensation	Oryn Delta	Low-visibility incident / fatigue / delayed rescue	2.53m

4.5 Cargo Loss

A detailed scenario testing plot has been provided in *Appendix E.3.2*. This indicates that the current product performs relatively well under optimal and moderate conditions, whilst the 1-in-100 worst case scenario indicates that the product cannot withstand catastrophic events on its own. This provides a useful guide for capital reserves required to survive severe shocks. The required capital can be split into two categories: reserves for day-to-day operations and reserves for extreme events

(see *Appendix E.3.3*). The capital held for regular operations at the 99% VaR level is Đ7.2B, but for extreme events is approximately Đ154.4B. The regular business reserves are required to be more liquid due to on demand needs. Meanwhile the second type of reserves can be held as a more strategic buffer to preserve solvency for 1-in-100 events. This recommendation acts as a continuation of GGI's already conservative fiscal policy.

Table 4.4 CL Top threats

Threat	Justification
Correlated Events	Due to affecting multiple variables, it can have complicated flow on effects. So, it is the unpredictability and uncertainty that make this the biggest threat to business operations.
Route Risk	Causes the largest increase in aggregate loss compared to any other single factors. Used in risk segregation and can potentially undermine pricing structure.
Solar Radiation & Debris Density	Both 'Solar Radiation' and 'Debris Density' carry a similar level of loss. However, solar flares are deemed a slightly greater threat to our model, as a spike in radiation can interfere with communication systems between ships and docks, as well as navigational programs. In contrast, 'Density Debris' is mainly a factor which impacts cargo only during transit, which self-healing hull technology can mitigate to an extent.

4.6 Risk Differences

Operations in these planetary systems carry many of the same risks, but to differing extents. Work in the Helionis Cluster is often exposed to erratic gravitational forces, causing micro-collisions and shifting debris clouds. Thus, the CL and EF products are likely to suffer the greatest losses in this planetary system, as these products are particularly susceptible to Debris risk. Similarly, the Zeta system also has dangerous space debris patterns and moderately sporadic solar flares, hence having a high Debris risk and Solar Flare risk, potentially impacting CL and WC to a greater degree. Finally, the Epsilon system carries abnormally large extraction risks due to severe weather and thermal patterns. This system therefore carries greater risks to the EF and BI products, as suggested by the threat tables.

Additionally, it is important to also consider the operating environments of other solar systems in which CQMC might eventually expand into. The Bayesian System is known for its above average radiation and temperature levels, which can create a habitat risk, suggesting greater claims for WC. However, this system generally is predictable in terms of radiation and space debris, so these risks are relatively lower for this operation. The Oryn Delta system is known for its sporadic solar flare and low luminosity, as well as its hazardous asymmetric asteroid ring. For this reason, expanding operations into this system is highly risky due to the Communication Failure risks, Debris risks and Radiation risks, which can greatly impact all four hazard products.

5. Assumptions

A major data-related assumption arose from the inconsistency between the data dictionaries and the actual data, with the value ranges/levels failing to match the given data. It was assumed that the values specified in the data dictionary were more trustworthy and were thus chosen as the basis for data cleaning. So, assumptions were made in order to reverse any systematic input errors. For example, the first crucial assumption made is that all negative numeric values were intended to be positive, where appropriate.

Further, Claim Amount, Cargo Value and Base Salary all had values which significantly exceeded their respective limits. For these variables it was assumed any value above the limit was written in cents rather than dollars and hence was divided by 100. In cases where the variables were below

their respective deductibles, they were disregarded. This assumption had a major impact on pricing for the way in which the majority of the raw claim amounts had to be shifted, impacting the scaling of pricing and capital and their projections. Furthermore, for Worker's Compensation, it was assumed that in the future, the Australian Labour Law regarding work-related injuries and compensation applied galaxy wide.

In cases of missing entries with the data sets, policy IDs were used to map missing values between the frequency and severity sheets where possible. If between the two sheets one had a value outside the defined range while the other was within it - the one in the limits was selected. If both were inside the limits, the conflicting rows would be removed. For other hazard specific assumptions see *Appendix B.4* (for BI) and *Appendix E.4* (for CL).

6. Data and Data Limitations

Data Limitations

While the data provided is expansive, it contained significant limitations that hindered robust product development. A major data limitation was the aforementioned inconsistency between the value ranges/levels provided in the data dictionaries and that of the actual data. This poses a major threat to our recommendations – if the data dictionaries are incorrect, the severity of every hazard coverage may have been underestimated. Thus the recommended premiums would not be high enough to cover actual costs, rendering the products infeasible.

The historical inflation and interest data is a limitation given its small 15-year period, causing difficulties in understanding true economic patterns. This limits the accuracy of long-term predictions, increasing variability in the models. Additionally, lacking historical environmental data, such as the weather patterns, limit capabilities for predicting claims and hinder understanding of the operating environments. The recommended products assume a stagnated view of the intergalactic mining industry as the models lack the capacity to incorporate effects of technological advancements upon future claim behaviours and additional costs.

Additionally, since the reason for each claim was not provided, the BI coverage triggers and exclusions could not be perfectly tailored to the nature of the claims, thus they have been kept relatively general. This poses a challenge as we are unaware of any common events that could provide great insight into how triggers and exclusions are set to favour net revenue. CL Cover specifically is limited in its success as a product, due to it being unable to provide specific solar system insights. As a result, it could be argued that this product serves more as a guide or temporary solution until we receive access to higher quality data.

Finally, all the aforementioned risks are related to specific operational and environmental issues which might arise in CQMC's practice. Therefore, it should also be addressed how the insurer will manage its legal risks, especially given CQMC's history of court issued fines. For example, claims procedures may still differ across countries or even planets and solar systems, so it is currently unclear which law would take precedence. This applies to dispute frameworks and laws which hold us legally liable to a claim. Our limited understanding of how laws operate on an intergalactic scale, impacts our ability to provide recommendations which could be accurately and practically applied.

Artificial Intelligence Usage

Appendix F details the usage of Artificial Intelligence in the completion of this project.

References

All data and supplementary references files are available on the 2026 SOA Research Institute Student Research Case Study Challenge webpage.

No external sources were utilised.

Appendix A – Project Objectives Summaries

Project Objectives Summary Tables:

***Table A.1*: Business Interruption**

Business Interruption Summary Data for Project Objectives								
Type	**Horizon**	**Mean**	**Variance**	**P5**	**P95**	**P99**	**TVaR95**	**TVaR99**
Cost	Short-Term	Đ652,000,000	9.62E+13	Đ636,000,000	Đ668,000,000	Đ675,000,000	Đ672,000,000	Đ679,000,000
	Long-Term	Đ3,510,000,000	9.57E+15	Đ3,370,000,000	Đ3,690,000,000	Đ3,780,000,000	Đ3,750,000,000	Đ3,830,000,000
Return	Short-Term	Đ730,000,000	1.81E+14	Đ718,000,000	Đ756,000,000	Đ756,000,000	Đ756,000,000	Đ756,000,000
	Long-Term	Đ3,940,000,000	1.74E+16	Đ3,750,000,000	Đ4,180,000,000	Đ4,300,000,000	Đ4,250,000,000	Đ4,350,000,000
Net Revenue	Short-Term	Đ77,711,642	2.79E+14	Đ54,107,514	Đ109,000,000	Đ119,000,000	Đ115,000,000	Đ124,000,000
	Long-Term	Đ428,000,000	1.09E+16	Đ274,000,000	Đ621,000,000	Đ705,000,000	Đ671,000,000	Đ737,000,000

***Table A.2*: Cargo Loss**

Aggregate Loss		
	Short Term (Đ)	*Long Term (Đ)*
Ranges (10% CI)	230 509 047 790 - 241 881 721 719	218 973 227 241 - 229 795 358 947
Expected Value	236 149 217 507	224 377 643 350
SD	3 583 477 293	3 431 153 109
Aggregate Returns		
Ranges (10% CI)	259343423269	246 585 957 385
Expected Value	259343423269	246585957385
SD	0	0
Aggregate Net Revenue		
Ranges (10% CI)	17461701550 - 28834375479	14537249959 - 23903156652
Expected Value	23194205762	22208314036
SD	3583477293	2973395165
General The long-term results have been averaged across 10 years. For this reason, they appear stable and similar to the short-term results, but are actually significantly more dispersed, evident when looking at the individual years and in the decreased expected profits.		
Tail Behaviours	Stable and relatively predictable	Becomes extreme and highly dispersed

***Table A.3*: Equipment Failure**

Type	Horizon	Ranges (10% CI)	Expected Value	Variance	SD	P5	P95	P99	TVaR95	TVaR99	Tail Behaviours
Aggregate Loss	Long term	444,706,449 - 485,822,337	465,200,307	1.56E+14	12,506,748	444,706,449	485,822,337	494,812,703	491,315,481	499,722,347	Stable and relatively predictable
Aggregate Loss	Short term	247,672,827 - 270,542,617	259,070,540	4.84E+13	6,959,638	247,672,827	270,542,617	275,523,266	273,606,738	278,276,859	Stable and relatively predictable
Aggregate Returns	Long term	708,952,965 - 708,952,965	708,952,965	0	0	708,952,965	708,952,965	708,952,965	708,952,965	708,952,965	Deterministic / no simulated tail
Aggregate Returns	Short term	394,125,852 - 394,125,852	394,125,852	0	0	394,125,852	394,125,852	394,125,852	394,125,852	394,125,852	Deterministic / no simulated tail
Aggregate Net Revenue	Long term	98,692,659 - 139,808,547	119,314,689	1.56E+14	12,506,748	98,692,659	139,808,547	89,702,293	93,199,515	84,792,649	Stable downside, remains positive
Aggregate Net Revenue	Short term	54,283,213 - 77,153,003	65,755,290	4.84E+13	6,959,638	54,283,213	77,153,003	49,302,564	51,219,091	46,548,971	Stable downside, remains positive

***Table A.4*: Workers' Compensation Summary**

Type	Horizon	Ranges (10% CI)	Expected Value	Variance	SD	P5	P95	P99	TVaR95	TVaR99	Tail Behaviours
Aggregate Loss	Long term	13,479,359 - 16,411,737	14,808,542	8.381E+11	915,466	13,479,359	16,411,737	17,475,642	17,085,879	18,179,224	Moderately heavy upper tail
Aggregate Loss	Short term	7,633,130 - 9,290,420	8,385,229	2.662E+11	515,935	7,633,130	9,290,420	9,873,273	9,660,686	10,266,792	Moderately heavy upper tail
Aggregate Returns	Long term	22,580,578 - 22,580,578	22,580,578	0	0	22,580,578	22,580,578	22,580,578	22,580,578	22,580,578	Deterministic / no simulated tail
Aggregate Returns	Short term	12,763,355 - 12,763,355	12,763,355	0	0	12,763,355	12,763,355	12,763,355	12,763,355	12,763,355	Deterministic / no simulated tail
Aggregate Net Revenue	Long term	2,205,417 - 5,137,794	3,808,612	8.381E+11	915,466	2,205,417	5,137,794	1,141,512	1,531,275	437,929	Stable downside, remains positive

Type	Horizon	Ranges (10% CI)	Expected Value	Variance	SD	P5	P95	P99	TVaR95	TVaR99	Tail Behaviours
Aggregate Net Revenue	Short term	1,228,725 - 2,886,015	2,133,917	2.662E+11	515,935	1,228,725	2,886,015	645,873	858,460	252,353	Stable downside, remains positive

Appendix B – Business Interruption

B.1 Product Design

For Business Interruptions coverage, we will provide 3 insurance products segmented based upon risk characteristics. The packages consist of: Core BI Protect, Enhanced BI Plus, and Catastrophic BI Ultimate. Premiums are subject to a range of individual factors and thus are able to differ between policies given the capabilities of our strong Q-RISK engine. The associated annual pure premium ranges can be visualised in *Figure B.1* and *Table B.1*.

***Figure B.1*: BI Pure Premium Ranges by Product Level**

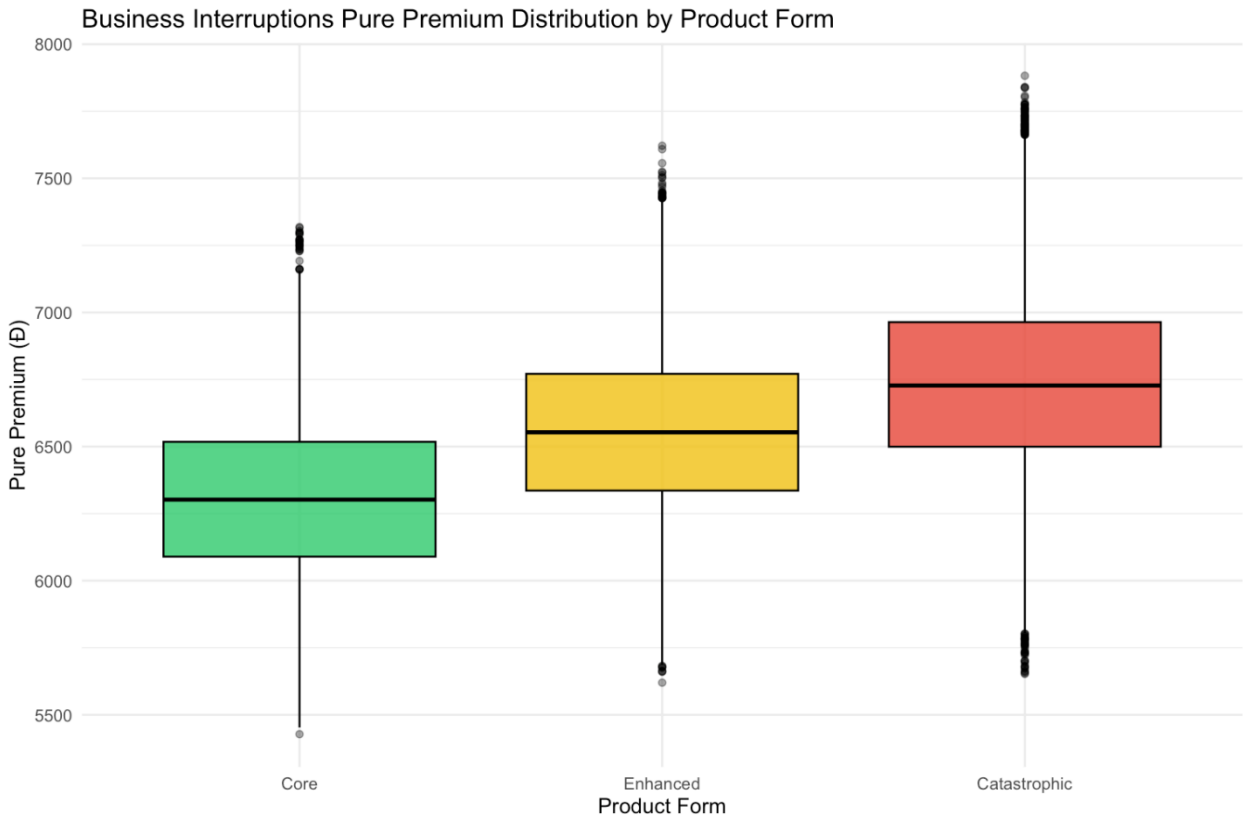


***Table B.1:* BI Pure Premium Ranges by Product Level**

Product	n policies	mean	P25	P50	P75	P95
Core	33265	Đ6,305.54	Đ6,089.58	Đ6,302.18	Đ6,517.53	Đ6,801.75
Enhanced	33264	Đ6,557.03	Đ6,335.83	Đ6,553.20	Đ6,771.04	Đ7,062.31
Catastrophic	33265	Đ6,732.35	Đ6,499.51	Đ6,727.78	Đ6,963.99	Đ7,282.79

Location based risk levels can be visualised in *Figure B.2* and *Figure B.3* for product splits by Station and System, respectively.

***Figure B.2*: BI Product Form by Station**

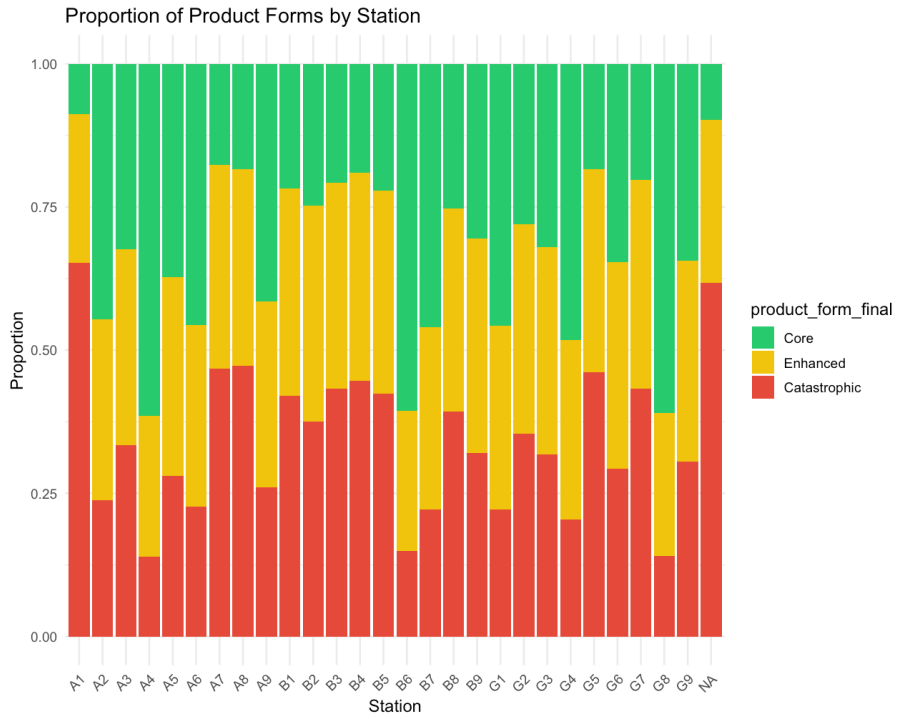
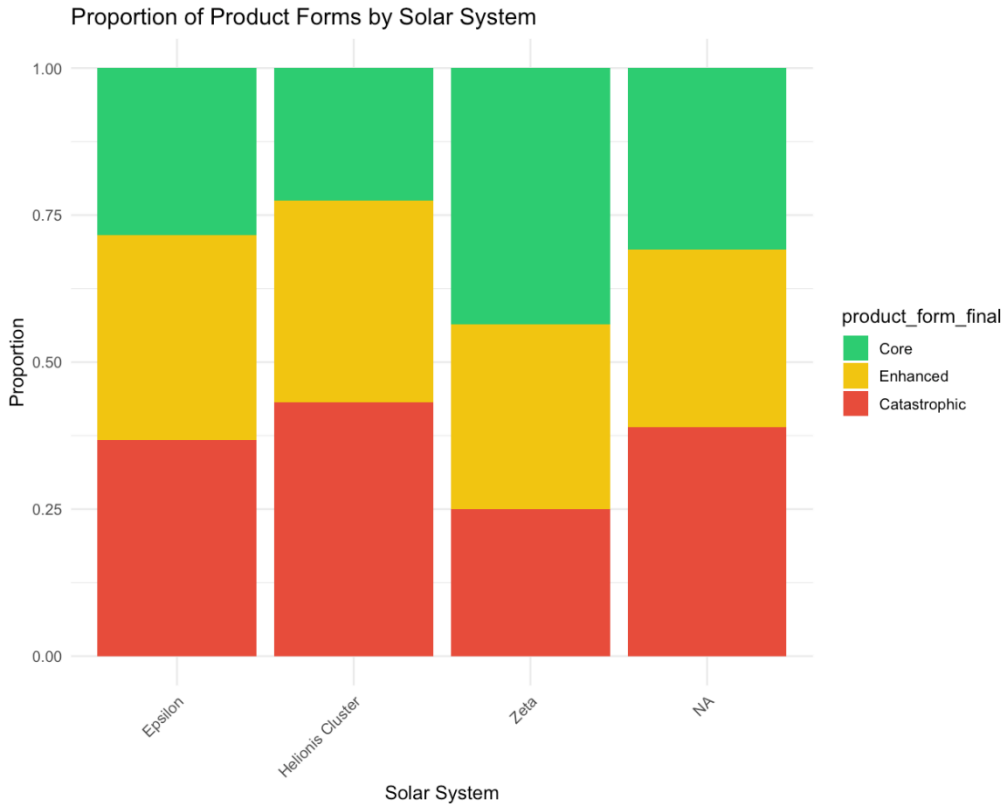


***Figure B.3*: BI Product Form by System**



B.1.1 Benefit Structure

i. Core BI Protect (Low Risk)

Core BI Protect is the most affordable BI product of the three, offering an average annual pure premium of Đ6306. This option is designed for strong operation resilience, informing crews of higher experience that prioritise high safety and low production loads. In terms of location, this product form targets stations of lower severity, such as A4, B6, G8 (see *Figure B.2*), and more stable solar systems, such as Zeta (See *Figure B.3*). Given the low-risk nature of this product, coverage is more basic but more affordable, thus acting as our volume product.

ii. Enhanced BI Plus (Medium Risk)

Enhanced BI Plus offers an average annual premium of Đ6557, aligning itself as a middle ground between our low and high risk BI products. This product has been designed for crews of moderate experience gaps, coming from more balanced stations, such as A3, B2, G2, and from solar systems of moderate volatility, much alike Epsilon. By selecting the Enhanced BI Plus product, coverage will be more advanced and more flexible, balancing coverage and risk control.

iii. Catastrophic BI Ultimate (High Risk)

The Catastrophic BI Ultimate product offers the highest annual premium, averaging Đ6732, yielding the highest risk coverage suitable for more extreme crises. This product is targeted at crews of low experience, prone to overloading production and having low safety. Additionally, it is perfect for more volatile systems, such as Helionis Cluster, and high risk stations, especially A1, but also other such as A7, A8, B3-B5, G5, and G7. This product focuses on protecting against high BI losses, capturing the tail-risk.

B.1.2 Triggers

i. Core BI Protect

Since this package is the low-risk option, this will have the lowest coverage thus we only pay when losses from business interruptions are clear, direct, and measurable. Triggers include physical damage to the insured asset that causes a full operational shutdown, measurable interruptions that halt production/output, and if forced business shutdown exceeds the minimum waiting period threshold of 48 hours. For this coverage level, only localised station-level incidents will be covered, thus reducing risk by removing dependency on external systems.

Core BI Protect coverage will NOT be triggered by any partial performance declines, upstream or downstream system failures, or gradual degradation and deterioration of assets. Additionally, thus coverage level will not extend to high-risk stations, including A1, A7, B3-B5, G5, and G7.

ii. Enhanced BI Plus

This is the medium-risk tier coverage choice, including everything in the Core package plus additional triggers. These additional triggers include partial interruptions, activated when output falls below 70%, and system inefficiencies, such as the underperformance of assets which impacts revenue. Additionally, basic coverage will be provided in the event that linked stations/suppliers within the same solar system fail and will have a shorter waiting period of 24 hours.

Enhanced BI Plus coverage will NOT be fully triggered by large-scale spill-on-effects from system collapses or extreme multi-system failures.

iii. Catastrophic BI Ultimate

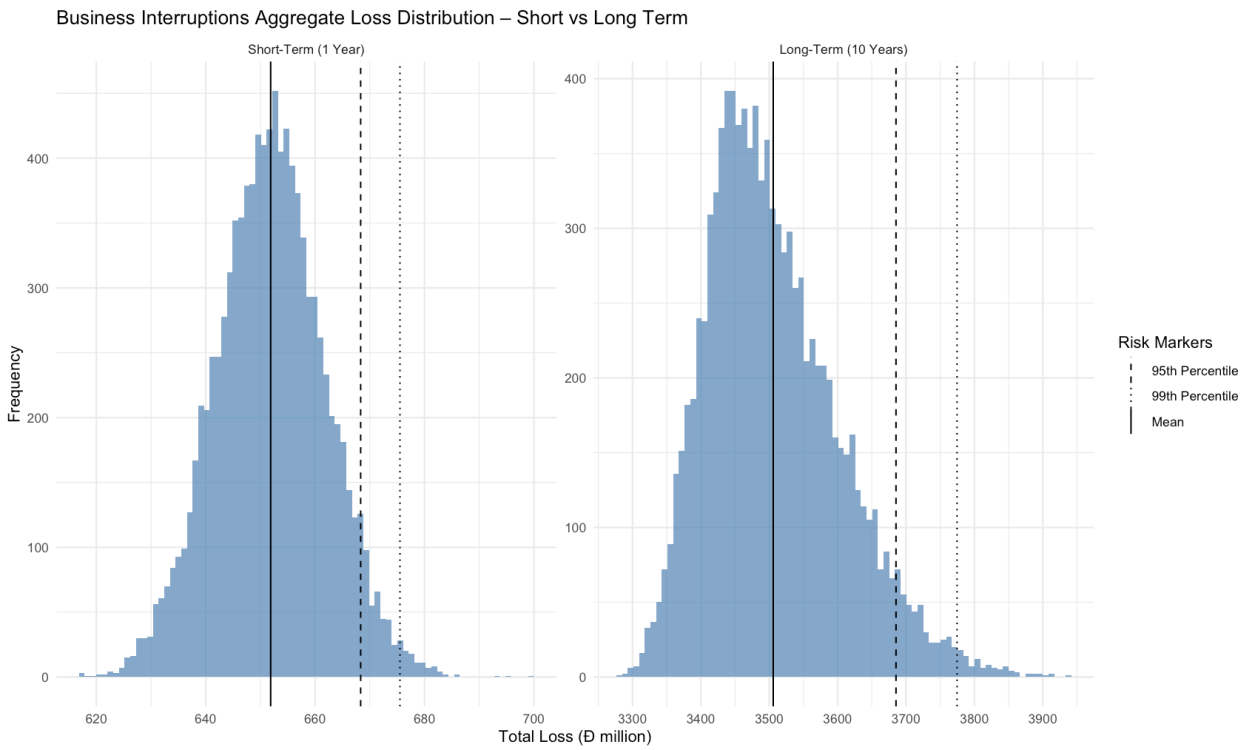
As the most advanced risk tier choice, Catastrophic BI Ultimate coverage will include everything in the Enhanced BI Plus package and more. It will cover against full system-wide failures, due to outages at solar system level, cascading chain reaction failures, and interdependency breakdowns, such as a critical supplier or network goes down. Additionally, this will cover disruptions of long durations for when shutdown periods extend beyond expectations and will entail no waiting time for major events.

B.2 Pricing & Capital Modelling

B.2.1 Aggregate Loss Distribution

The short and long-term Aggregate Loss Distributions for Business Interruptions have been presented in *Figure B.4*.

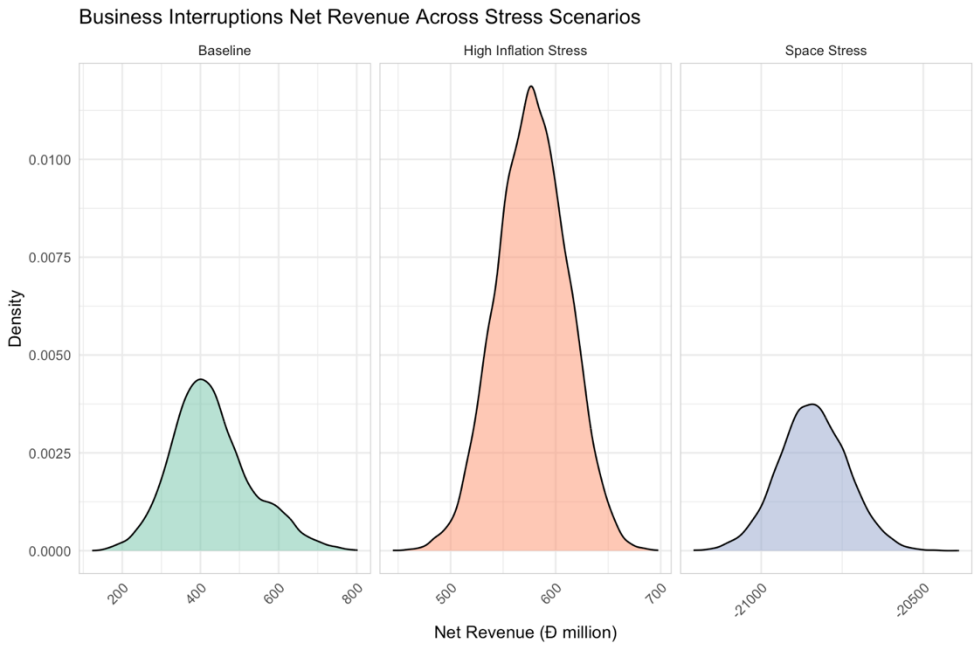
***Figure B.4*: BI Aggregate Loss Distributions**



B.2.2 Stress Testing

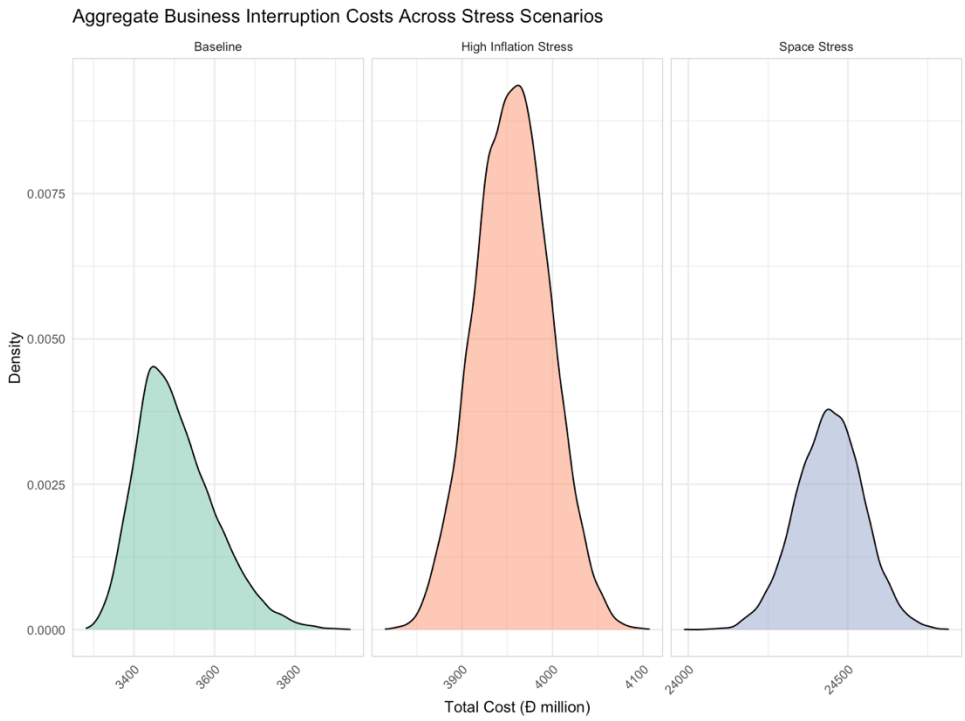
The BI Net Revenue under stress testing scenarios can be visualised in *Figure B.5*.

***Figure B.5*: BI Stress Testing Scenarios – Net Revenue**



The BI Costs under stress testing scenarios can be visualised in *Figure B.6*.

***Figure B.6*: BI Stress Testing Scenarios – Costs**

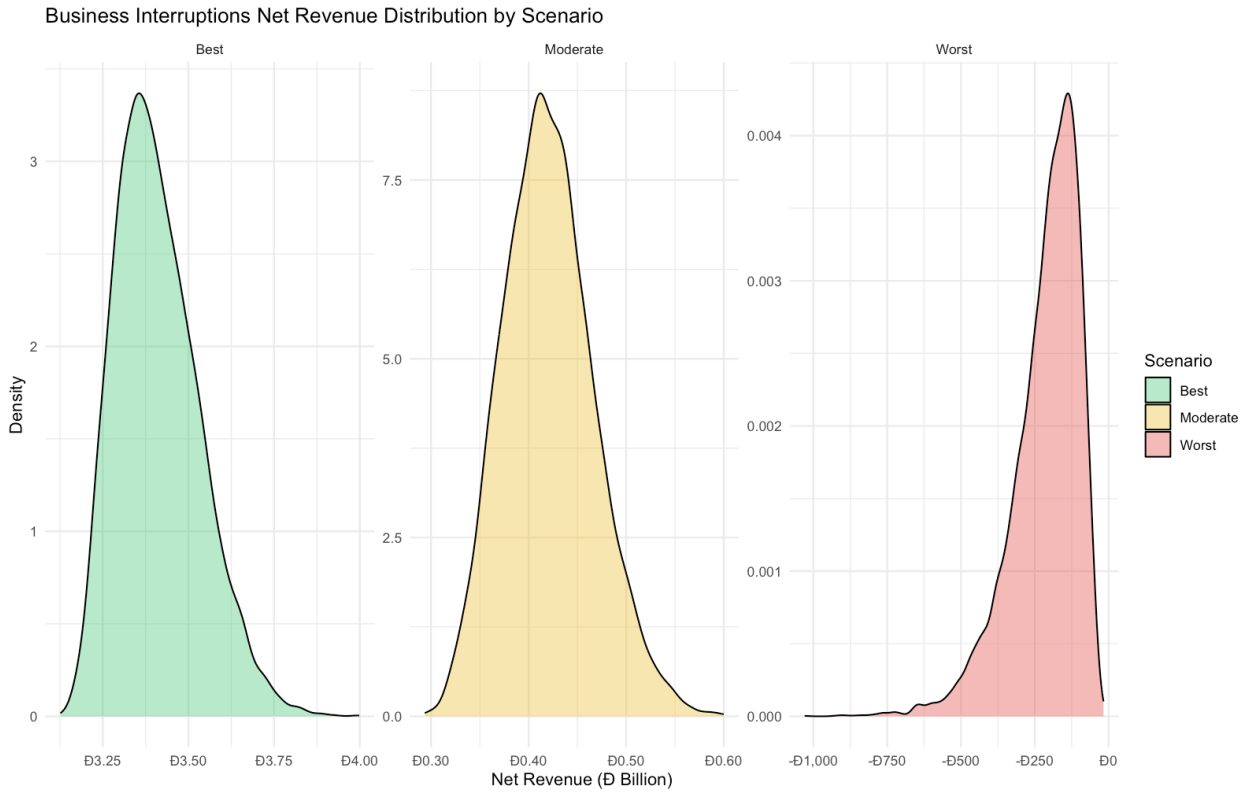


B.3 Risk Assessment

B.3.1 Scenario Testing

The BI Net Revenue under Best/Moderate/Worst case Scenario Testing scenarios can be visualised in *Figure B.7*.

***Figure B.7*: BI Scenario Testing Scenarios – Net Revenue**



B.4 Assumptions

Given the data dictionary versus dataset inconsistencies, BI data ranges had to be adjusted for variables including production load, energy backup score, supply chain index, average crew experience, maintenance frequency , safety compliance, exposure, and claim count. For these features:

- If the values were above the set range, they were capped to the maximum defined value.
- If the values were below the set range, they were capped to the minimum defined value.
- If a variable was supposed to be an integer, for example if it was of the category type, all non-integer entries were rounded to their nearest integer.

B.5 Modelling

The final BI frequency model was a logit model, and its summary has been included below. This model was used to predict the probability of a claim occurring, p_i, given the provided variables.

```
Call:
glm(formula = I(claim_count_clean > 0) ~ safety_compliance_clean +
    energy_backup_score_clean + production_load_clean + avg_crew_exp_clean +
    solar_system_clean, family = binomial(), data = freq_model_data)

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                      -2.623002   0.052332 -50.122  < 2e-16 ***
safety_compliance_clean           0.005906   0.008782   0.673  0.50122
energy_backup_score_clean         0.025625   0.008766   2.923  0.00346 **
production_load_clean            -0.023111   0.042986  -0.538  0.59082
avg_crew_exp_clean               -0.001314   0.001431  -0.918  0.35845
solar_system_cleanHelionis Cluster -0.039534   0.034130  -1.158  0.24673
solar_system_cleanZeta           -0.022099   0.027661  -0.799  0.42434
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 50566  on 98995  degrees of freedom
Residual deviance: 50554  on 98989  degrees of freedom
AIC: 50568

Number of Fisher Scoring iterations: 5
```

The final BI severity model was a log-normal regression model, and its summary appears below. This model was used to predict the expected severity of a claim given that a claim was made, $E[\text{Severity}_i|\text{ Claim}]$.

```
Call:
lm(formula = log_claim ~ safety_compliance_clean + solar_system_clean +
    station_id_clean, data = sev_step_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.28426 -0.59228 -0.00392  0.57998  2.71345

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                      11.496452   0.040911 281.010   <2e-16 ***
safety_compliance_clean           0.005233   0.004881   1.072   0.2837
solar_system_cleanHelionis Cluster  0.013207   0.019222   0.687   0.4921
solar_system_cleanZeta           -0.031121   0.015437  -2.016   0.0438 *
station_id_cleanA2               -0.090211   0.053191  -1.696   0.0899 .
station_id_cleanA3               -0.067921   0.051020  -1.331   0.1831
station_id_cleanA4               -0.129277   0.053305  -2.425   0.0153 *
station_id_cleanA5               -0.079045   0.052491  -1.506   0.1321
station_id_cleanA6               -0.097068   0.052188  -1.860   0.0629 .
station_id_cleanA7               -0.034743   0.053526  -0.649   0.5163
station_id_cleanA8               -0.035725   0.050946  -0.701   0.4832
station_id_cleanA9               -0.086650   0.051271  -1.690   0.0911 .
station_id_cleanB1               -0.046336   0.050254  -0.922   0.3565
station_id_cleanB2               -0.055498   0.049751  -1.115   0.2647
station_id_cleanB3               -0.041143   0.051716  -0.796   0.4263
station_id_cleanB4               -0.036191   0.051340  -0.705   0.4809
station_id_cleanB5               -0.045532   0.051422  -0.885   0.3759
station_id_cleanB6               -0.125917   0.051853  -2.428   0.0152 *
station_id_cleanB7               -0.095657   0.051570  -1.855   0.0636 .
station_id_cleanB8               -0.054829   0.052036  -1.054   0.2921
station_id_cleanB9               -0.065663   0.052854  -1.242   0.2141
station_id_cleanG1               -0.090806   0.050964  -1.782   0.0748 .
station_id_cleanG2               -0.063308   0.050929  -1.243   0.2139
station_id_cleanG3               -0.070268   0.051678  -1.360   0.1739
station_id_cleanG4               -0.099881   0.051127  -1.954   0.0508 .
station_id_cleanG5               -0.039577   0.052032  -0.761   0.4469
station_id_cleanG6               -0.072814   0.050458  -1.443   0.1490
station_id_cleanG7               -0.041614   0.052907  -0.787   0.4316
station_id_cleanG8               -0.126691   0.051704  -2.450   0.0143 *
station_id_cleanG9               -0.073082   0.051708  -1.413   0.1576
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6908 on 9943 degrees of freedom
Multiple R-squared:  0.00276,	Adjusted R-squared:  -0.0001481
F-statistic: 0.9491 on 29 and 9943 DF,  p-value: 0.5435
```

To arrive at the pure premium for each policy, the following formula was followed:

$$\text{Pure Premium} = p_i * E[\text{Severity}_i|\text{ Claim}].$$

Appendix C – Equipment Failure

C.1 Product Design

We offer three Equipment Failure forms because the machinery book is structurally uneven. The distribution of risk is shaped by machine class, operating environment, maintenance profile, and usage intensity. A single undifferentiated machinery product would force the safer asset pool to subsidise the tail, while still undercharging the largest and most severe risks. The three-form structure gives us room to align cover, premium, and underwriting terms with modelled risk instead of treating every machine as if it lives in the same operating world.

Equipment Failure Essentials is our lower-risk product and is intended for the bottom of the machinery loss distribution. It is the right form where losses are still meaningful but remain closer to attritional experience than to catastrophe-style tail events. Equipment Failure Core is our middle form and is designed for assets that clearly sit above the safer pool but do not dominate capital strain. Equipment Failure Plus is the upper-tail form. It is built for machine classes where breakdown severity, concentration, or environmental stress make the tail materially wider and the need for stronger terms much more obvious.

The numerical ladder confirms why the product is split this way. The charged premium cut-offs run from Đ18,710 to Đ28,886 in the Very Low tier, Đ28,886 to Đ32,217 in the Low tier, Đ32,217 to Đ45,682 in the Standard tier, Đ45,682 to Đ62,478 in the High tier, and Đ62,478 to Đ157,849 in the Very High tier. That spread is too wide for a single machinery form to remain credible. The three-form structure is therefore not just packaging. It is a response to the shape of the underlying loss model.

The underwriting terms strengthen with risk. Deductibles rise, coinsurance appears in the higher tiers, and waiting periods shorten as the machinery becomes more severe and more capital-intensive. This matters because risk control in Equipment Failure does not come from premium alone. Premium signals price, but deductibles and coinsurance also keep CQMC sharing in the most severe losses and reduce the risk that upper-tier machinery is effectively written on attritional terms.

Table C.1.1 Equipment Failure product ladder

Product form	Risk tiers	Charged premium band	Typical underwriting position
Equipment Failure Essentials	Very Low	Đ18,710 to Đ28,886	Lower deductibles, no coinsurance, longer waiting period
Equipment Failure Core	Low to Standard	Đ28,886 to Đ45,682	Moderate deductibles, no coinsurance, standard waiting period
Equipment Failure Plus	High to Very High	Đ45,682 to Đ157,849	Higher deductibles, coinsurance, shorter waiting period

Table C.1.2 Equipment Failure machine-level premium schedule (full table)

Solar system	Risk tier	Product form	Machine type	Units	Annual pure premium	Annual charged premium	Deductible	Coinsurance	Waiting period
Bayesia System	Very Low	Essentials	Mag-Lift Aggregator	150	25,217	27,739	5%	0%	8 hrs
Bayesia System	Low	Core	Ion Pulverizer	45	28,738	31,612	7%	0%	8 hrs
Bayesia System	Low	Core	FluxStream Carrier	75	28,204	31,024	4%	0%	8 hrs
Bayesia System	Standard	Core	Graviton Extractor	120	39,128	43,040	9%	0%	8 hrs
Bayesia System	Standard	Core	Flux Rider	750	36,751	40,426	8%	0%	8 hrs
Bayesia System	High	Plus	Quantum Bore	150	53,486	58,835	11%	10%	8 hrs
Helionis Cluster	High	Plus	Mag-Lift Aggregator	300	55,158	60,674	9%	10%	6 hrs
Helionis Cluster	High	Plus	Ion Pulverizer	90	48,745	53,620	10%	10%	6 hrs
Helionis Cluster	Very High	Plus	Quantum Bore	300	143,499	157,849	13%	15%	6 hrs
Helionis Cluster	Very High	Plus	Graviton Extractor	240	91,820	101,002	13%	15%	6 hrs
Helionis Cluster	Very High	Plus	FluxStream Carrier	150	80,249	88,274	10%	15%	6 hrs
Helionis Cluster	Very High	Plus	Flux Rider	1,500	57,891	63,680	12%	15%	6 hrs
Oryn Delta	Very Low	Essentials	Ion Pulverizer	30	25,285	27,813	7%	0%	12 hrs
Oryn Delta	Very Low	Essentials	Flux Rider	500	24,501	26,951	4%	0%	12 hrs
Oryn Delta	Very Low	Essentials	Mag-Lift Aggregator	100	17,009	18,710	4%	0%	12 hrs
Oryn Delta	Low	Core	FluxStream Carrier	50	27,723	30,495	5%	0%	12 hrs
Oryn Delta	Standard	Core	Quantum Bore	100	39,726	43,698	9%	0%	12 hrs

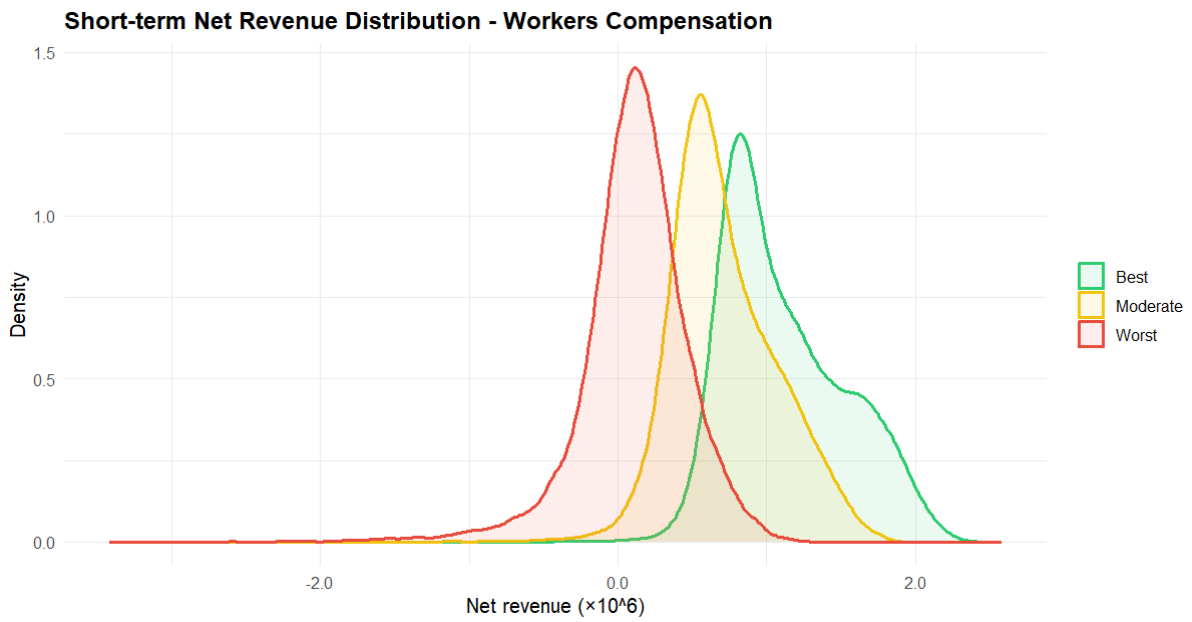
Oryn Delta	Standard	Core	Graviton Extractor	80	29,425	32,368	9%	0%	12 hrs

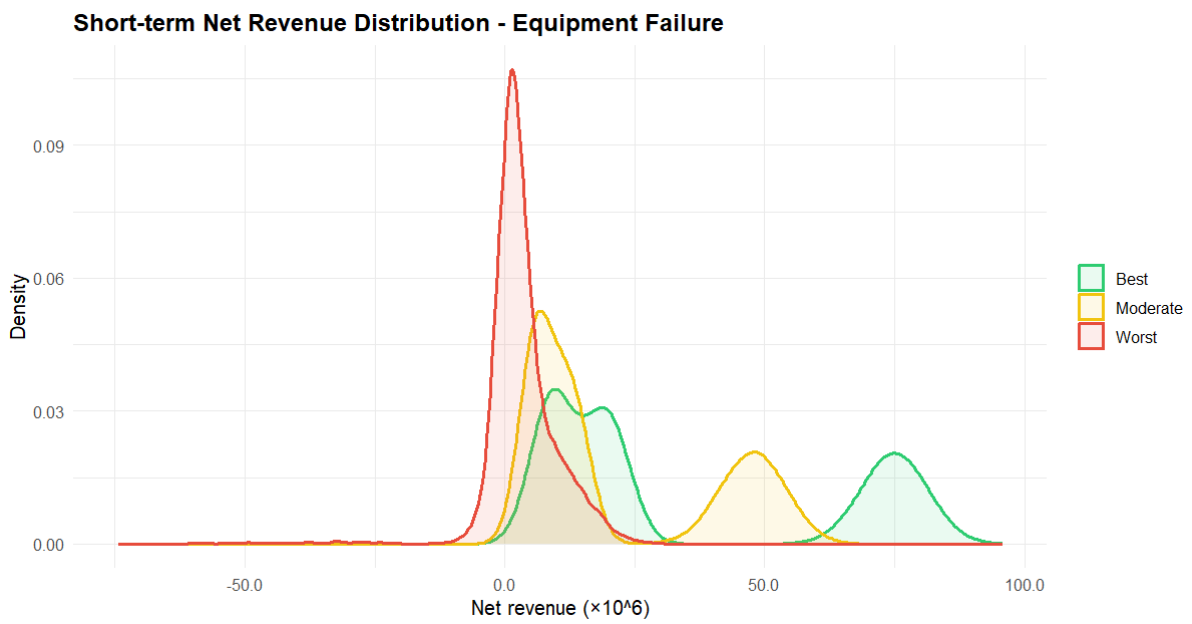
Table C.1.3 Equipment Failure risk-tier charged premium cut-offs

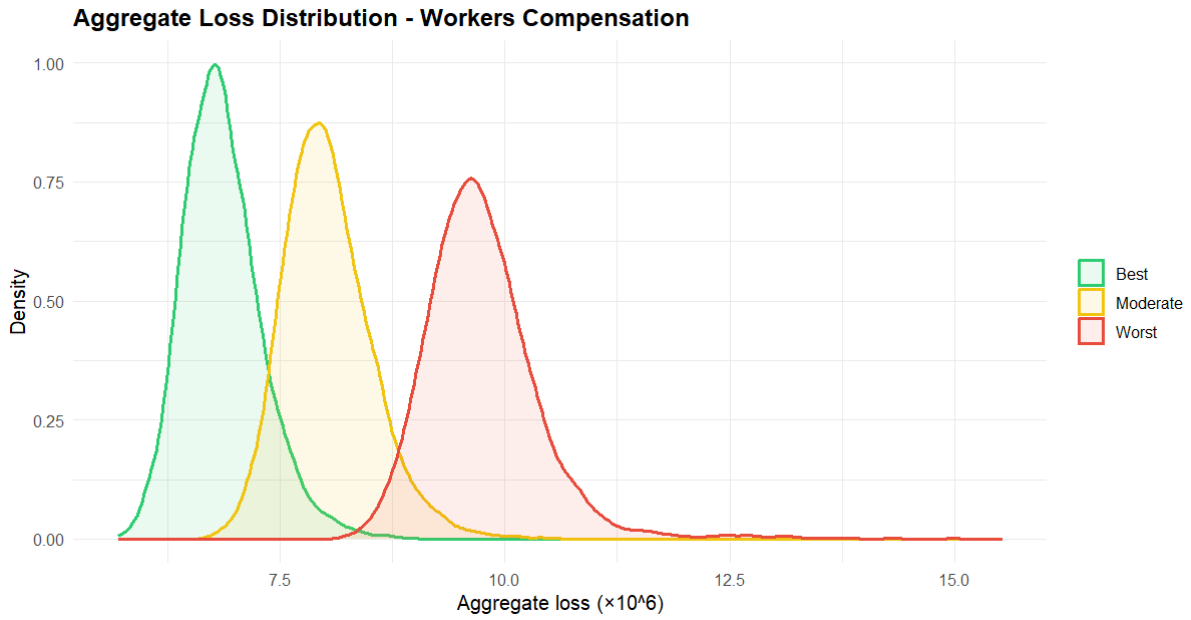
Risk tier	Charged premium lower bound	Charged premium upper bound
Very Low	18,710	28,886
Low	28,886	32,217
Standard	32,217	45,682
High	45,682	62,478
Very High	62,478	157,849

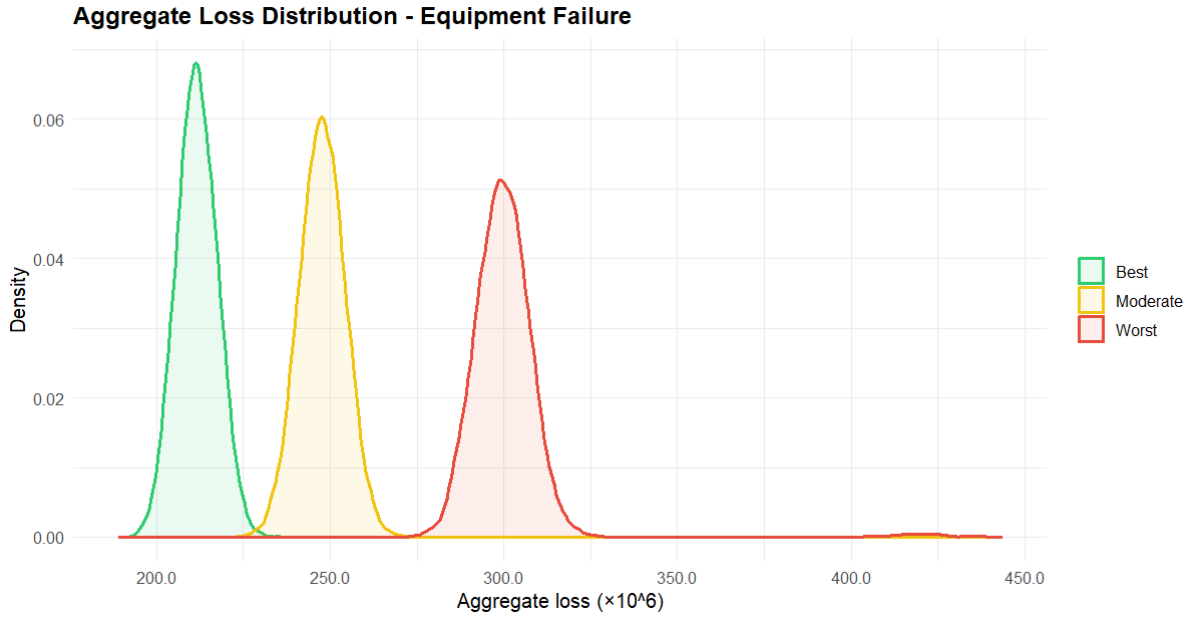
The full schedule makes the segmentation concrete. Bayesia's ladder moves from a Mag-Lift Aggregator at Ð27,739 up to a Quantum Bore at Ð58,835. Helionis is much more capital-heavy, with a Quantum Bore at Ð157,849 and a Graviton Extractor at Ð101,002. Oryn Delta sits lower on average but still contains meaningful middle-tier assets such as a Quantum Bore at Ð43,698. The schedule therefore supports three distinct commercial stances: Helionis as the scale market, Bayesia as an environmentally loaded market, and Oryn as a more conservatively underwritten market where fragility rather than average premium is the key concern.

C.2 Pricing and Capital Model









C.2.1 - Feature Engineering & Variable Definitions

The Equipment Failure model begins with the core operational and claims variables: equipment age, maintenance interval, usage intensity, exposure, claim count, claim amount, solar system status, and equipment type. Because equipment risk is unlikely to vary linearly with age, usage, or servicing practices, these raw inputs are transformed into bands, nonlinear terms, and interactions that better capture how deterioration builds in practice.

The engineered variables include age_band, maint_band, usage_band, maintenance_k, wear_score, maint_load, maint_inverse, age_sq, usage_sq, age_x_usage, usage_x_maint, and annual_claim_rate. Together, these variables are designed to reflect key operational risk mechanisms. For example, wear_score and age_x_usage capture the greater failure risk associated

with older equipment operating under heavier usage, while usage_x_maint captures the added strain created when intensive use is paired with less frequent servicing. maint_inverse allows shorter maintenance intervals to appear as a protective effect in the model, rather than only representing risk through longer servicing gaps.

These engineered features are important not only for pricing but also for scenario analysis. A maintenance backlog shock increases maintenance intervals, which in turn affects maintenance_k, maint_load, maint_inverse, and usage_x_maint, transmitting the shock through several parts of the model. Similarly, a usage shock raises usage intensity and flows through wear_score, usage_sq, age_x_usage, and usage_x_maint, allowing the stress test to reflect both direct and compounding effects on equipment failure risk.

C.2.2 - Further Explanation

We price Equipment Failure as the dominant earnings and capital line in the package. In the short term, expected annual loss is 247.55m, projected claims cost is 259.00m, charged premium is 385.0m, and annual net revenue is 65.8m. In the long term, expected annual loss remains 247.55m, projected claims cost rises to 465.07m, charged premium rises to 691.0m, and annual net revenue rises to 119.0m. Expected loss is shown as a positive cost because it is an expected claims outflow. It becomes a negative contribution only when it is subtracted inside the net revenue calculation.

The aggregate-loss results show a progressive worsening from Best to Moderate to Worst_1in100. In the short term, mean loss rises from 228m to 267m to 324m, P95 from 238m to 278m to 339m, P99 from 243m to 284m to 376m, and TVaR99 from 245m to 286m to 388m. In the long term, the same sequence widens materially: mean loss rises from 409m to 479m to 581m, P95 from 428m to 500m to 608m, P99 from 436m to 509m to 676m, and TVaR99 from 440m to 514m to 697m. The long-term view therefore compounds more than the mean; it expands the upper tail substantially.

The stress tests explain what drives that expansion. A maintenance backlog shock represents longer maintenance intervals and weaker reliability control. It increases mean loss by 10% relative to the Moderate case. A usage intensity shock represents heavier utilisation, which increases wear load and the chance that a breakdown is both more frequent and more severe. It lifts mean loss by 15%. A catastrophe severity shock represents a more severe system-wide breakdown environment and increases mean loss by 35%. The combined operating shock applies all of these forces together and raises mean loss by 50%. This is the numerical reason we reserve the Plus form for the upper tail: not only are losses larger, they become more nonlinear under operational deterioration.

Dependency is built in through the common-shock catastrophe structure. The Worst_1in100 case is not merely a scaled-up independent loss. It is a correlated multi-system event. Moving from Moderate to Worst_1in100 increases short-term mean loss by 21.4%, short-term P99 by 32.7%, and short-term TVaR99 by 35.5%. The long-term uplifts are 21.4%, 32.8%, and 35.7% respectively. This shows that correlation matters most in the tail, which is exactly where capital discipline matters most.

C.2.3 - Methodology

Best uses 0.90x frequency, 0.95x severity, and no shared event. Moderate uses 1.00x frequency, 1.00x severity, and no shared event. Worst_1in100 uses 1.10x frequency, 1.10x severity, and a 1% shared-event overlay. Short-term mean loss moves from 228m to 267m to 324m. Long-term mean loss moves from 409m to 479m to 581m.

The stress tests change specific variables. Maintenance backlog shock increases maintenance_int, which also increases maintenance_k and usage_x_maint and reduces maint_inverse. Usage intensity shock increases usage_intensity, which also increases wear_score, usage_sq, age_x_usage, and usage_x_maint. Catastrophe severity shock leaves attritional settings unchanged and increases catastrophe severity after the shared event. Combined operating shock applies all three.

Table C.2.1 Equipment Failure pricing and capital summary

Horizon	Expected annual loss (m)	Projected claims cost (m)	Charged premium (m)	Annual net revenue (m)	Moderate mean loss (m)	Moderate P99 (m)	Worst-case TVaR99 (m)
Short term	247.55	259.00	385.0	65.8	267	284	388
Long term	247.55	465.07	691.0	119.0	479	509	697

Table C.2.2 Equipment Failure scenario definitions

Scenario	Definition	Interpretation in model
Best	Smooth operations with attritional claims only	Lower-frequency, lower-severity operating environment with no correlated catastrophe overlay
Moderate	Base operating conditions	Central case used for pricing, capital, and stress comparison
Worst_1in100	Correlated catastrophe with shared event	Multi-system common-shock event that loads frequency and severity simultaneously

Table C.2.3 Equipment Failure aggregate loss results

Horizon	Scenario	Mean loss (m)	P95 (m)	P99 (m)	TVaR99 (m)
Short term	Best	228	238	243	245
Short term	Moderate	267	278	284	286
Short term	Worst_1in100	324	339	376	388
Long term	Best	409	428	436	440
Long term	Moderate	479	500	509	514
Long term	Worst_1in100	581	608	676	697

Table C.2.4 Equipment Failure stress tests

Stress test	Horizon	Base mean loss (m)	Stressed mean loss (m)	Mean uplift	P95 (m)	P99 (m)	TVaR99 (m)
Maintenance backlog shock	Short term	267	293	+10%	306	312	315
Usage intensity shock	Short term	267	307	+15%	320	326	329
Catastrophe severity shock	Short term	267	360	+35%	376	383	386
Combined operating shock	Short term	267	400	+50%	417	425	429
Maintenance backlog shock	Long term	479	527	+10%	550	560	565
Usage intensity shock	Long term	479	551	+15%	575	585	591
Catastrophe severity shock	Long term	479	646	+35%	674	687	693
Combined operating shock	Long term	479	718	+50%	749	764	770

Table C.2.5 Equipment Failure dependency test

Horizon	Moderate mean loss (m)	Worst_1in100 mean loss (m)	Mean uplift	Moderate P99 (m)	Worst_1in100 P99 (m)	P99 uplift	Moderate TVaR99 (m)	Worst_1in100 TVaR99 (m)	TVaR99 uplift
Short term	267	324	21.4%	284	376	32.7%	286	388	35.5%
Long term	479	581	21.4%	509	676	32.8%	514	697	35.7%

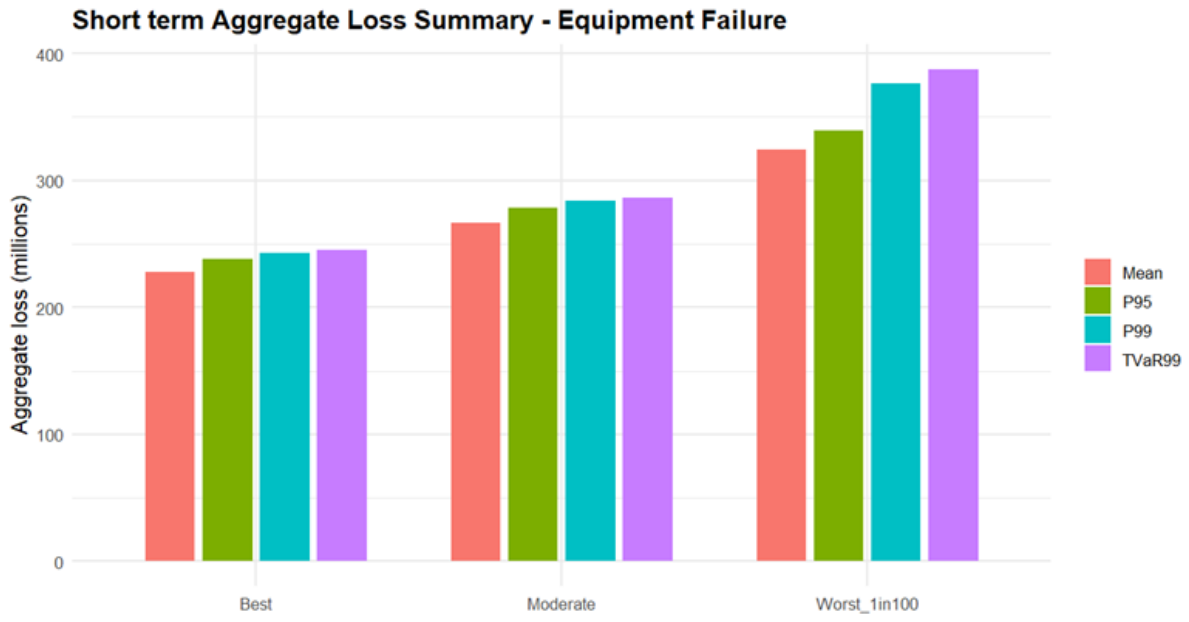


Figure C.2.1 Short-term Equipment Failure aggregate loss summary.

Figure C.2.1 shows that even in the short term, the distribution does not simply shift to the right as conditions worsen. It also spreads out, especially between P99 and TVaR99 in the worst case. That widening gap is a visual reminder that the tail is not behaving like an attritional line.

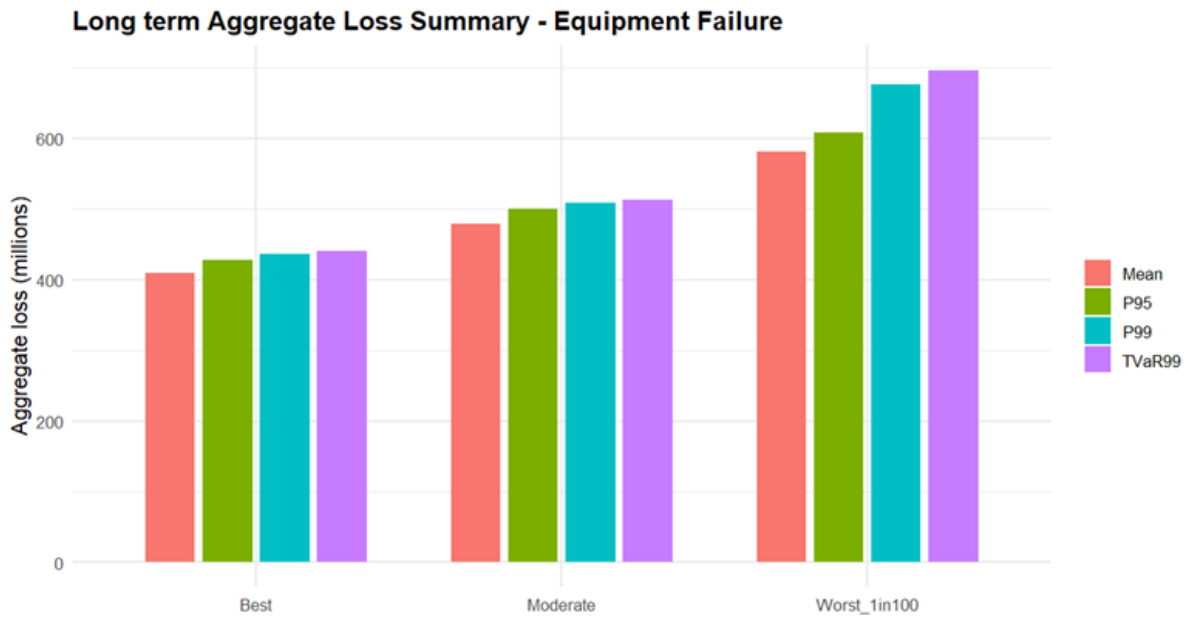


Figure C.2.2 Long-term Equipment Failure aggregate loss summary.

Figure C.2.2 shows the same pattern at a larger scale. The long-term worst-case bar stack is visually further away from the Best and Moderate cases than in the short-term figure, which is consistent with the numerical widening in P99 and TVaR99.

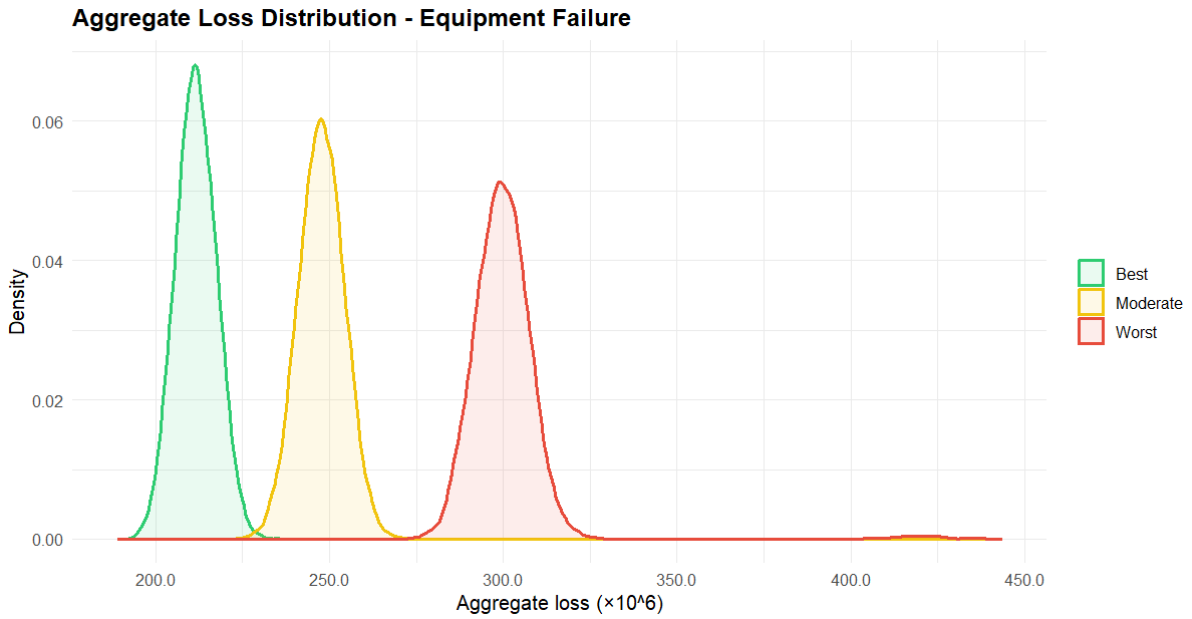


Figure C.2.3 Short-term Equipment Failure net revenue distribution by scenario.

Figure C.2.3 is useful because it links the loss analysis to commercial consequences. The Best case has a strongly positive net revenue centre, the Moderate case remains positive but much wider, and the Worst_1in100 case pushes mass much closer to zero and into the negative range.

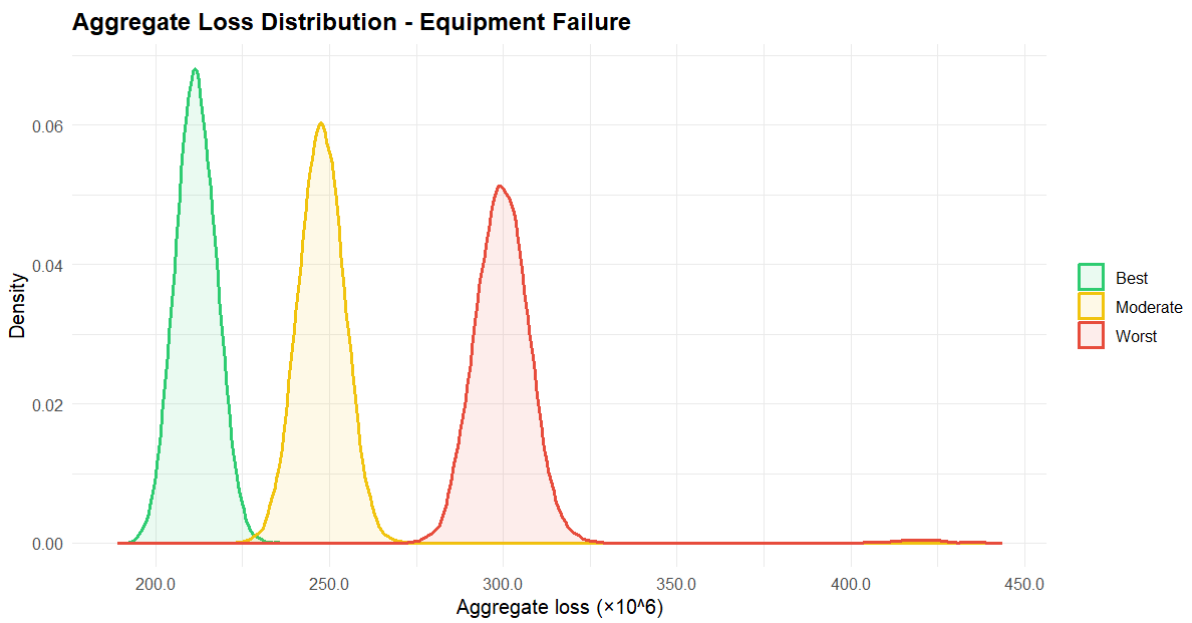


Figure C.2.4 Equipment Failure aggregate loss density by scenario.

Figure C.2.4 provides the distributional view behind the bar summaries. The rightward shift from Best to Moderate to Worst_1in100 is clear, but the more important feature is the fatter upper tail in the worst case.

C.3 Risk Assessment

C.3.1 - Feature Engineering & Variable Definition

The risk assessment view uses a simplified set of underwriting indicators derived from the pricing model, focusing on the main features most relevant for portfolio classification and scenario analysis. These include age_flag, usage_flag, maintenance_flag, risk_tier, and the solar-system peril mapping. age_flag groups fleets into younger, mid-life, and mature categories, usage_flag distinguishes moderate use from heavy and near-continuous operation, and maintenance_flag separates tight servicing schedules from moderate and wide maintenance gaps. Together, these indicators provide a more interpretable summary of underlying equipment risk than the full pricing specification.

The scenario overlays then stress the specific operating variables linked to these indicators. A maintenance backlog shock increases maintenance_int, which in turn affects maintenance_k, maint_inverse, and usage_x_maint. A usage intensity shock increases usage_intensity, flowing through wear_score, usage_sq, age_x_usage, and usage_x_maint. A catastrophe severity shock affects claim severity only when the shared_event flag is triggered, reflecting losses arising from common external events rather than routine operating conditions. The combined operating shock applies all three stresses simultaneously, allowing the model to capture the effect of multiple adverse conditions occurring at once.

C.3.2 - Further Explanation

Equipment Failure risk is not distributed evenly across the three solar systems. Helionis Cluster is the largest absolute exposure and the main concentration-risk market, with short-term expected annual loss of about 180.0m. Bayesia System is materially smaller at about 46.4m, but its risk is shaped by radiation spike, thermal shock, and shielding failure. Oryn Delta is smaller again at about 21.6m, yet remains the most fragile downside market because flare disruption, orbital shear, and relay outage can turn an otherwise moderate machine failure into a broader operational problem.

Correlated risk matters because the major space-weather and infrastructure hazards are not cleanly independent by solar system. A shared solar event or broader communications disruption can hit more than one system inside the same simulation. We therefore treat the worst case as a correlated catastrophe rather than a simple scalar increase in individual-system losses. This matters especially for Helionis because concentration and machine severity amplify the tail once a common shock occurs.

Scenario testing follows a clear three-case structure. The Best case assumes smooth operations with attritional claims only. The Moderate case is the base operating state used for pricing and capital comparison. The Worst_1in100 case assumes a correlated catastrophe with a shared-event overlay. Numerically, short-term mean aggregate loss moves from 228m in Best to 267m in Moderate and 324m in Worst_1in100, while long-term mean loss moves from 409m to 479m to 581m. The key point is not just that the average rises. It is that the tail widens at the same time.

The threat ranking is also very clear. The largest Equipment Failure threat is the combined operating shock, followed by catastrophe severity, usage intensity, and maintenance backlog. The

usage shock matters because it shows what 'heavier utilisation' means numerically: it pushes short-term mean loss from 267m to 307m and long-term mean loss from 479m to 551m. That is a material change from one operating driver. In practical terms, this is why we do not place high-use and low-use machinery on the same product form.

The solar-system narrative supports differentiated underwriting. Helionis is the scale market and the largest source of aggregate loss. Bayesia is technically priceable but demands explicit environmental loadings because of the radiation and thermal environment. Oryn Delta remains the system that should be underwritten most cautiously because its operational fragility pushes downside outcomes beyond what its average premium alone might suggest.

C.3.3 - Methodology

Scenario testing uses the same three states as the pricing model. Best uses 0.90x frequency, 0.95x severity, and no shared event. Moderate uses 1.00x frequency, 1.00x severity, and no shared event. Worst_1in100 uses 1.10x frequency, 1.10x severity, and a 1% shared-event overlay with system catastrophe multipliers. Short-term mean loss moves from 228m to 267m to 324m. Long-term mean loss moves from 409m to 479m to 581m.

The threat table maps directly to the variables. Maintenance backlog shock increases maintenance_int, maintenance_k, and usage_x_maint and reduces maint_inverse. Usage intensity shock increases usage_intensity, wear_score, usage_sq, age_x_usage, and usage_x_maint. Catastrophe severity shock increases catastrophe severity after the shared event only. Combined operating shock applies all three. Helionis is the main concentration market. Bayesia needs explicit radiation and thermal loadings. Oryn Delta needs the strongest terms because recovery is weaker and event amplification is harsher.

Table C.3.1 Equipment Failure top threats

Rank	Threat	Mean loss uplift vs Moderate
1	Combined operating shock	+50%
2	Catastrophe severity shock	+35%
3	Usage intensity shock	+15%
4	Maintenance backlog shock	+10%

Table C.3.2 Equipment Failure solar-system risk profile

Solar system	Primary threat	Short-term expected annual loss (m)
Helionis Cluster	Debris impact / micro-collision / communications clutter	180.0
Bayesia System	Radiation spike / thermal shock / shielding failure	46.4

Oryn Delta	Flare disruption / orbital shear / relay outage	21.6

Appendix D - Workers' Compensation

D.1 Product Design

We offer three Workers' Compensation forms because workforce cost is driven by more than payroll size. The model differentiates by role, employment type, salary, predicted claim duration, and operating environment. A flat payroll-based product would therefore ignore real differences in injury severity and recovery time. The three-form structure gives us a clean external package while retaining a detailed internal pricing ladder.

Workers Compensation ActiveCare Essentials covers the lower-risk workforce segments. These are roles that still generate genuine claims cost, but do so with more attritional loss behaviour and less duration strain. Workers Compensation ActiveCare covers the middle of the workforce distribution, where injury risk and recovery time are meaningful but not yet in the extreme tail. Workers Compensation ActiveCare Plus covers the upper-risk segments where claim severity, longer absences, or more volatile operating conditions make the cost of injury materially heavier.

The charged premium cut-offs reinforce this structure. The Very Low tier runs from Đ62 to Đ147, the Low tier from Đ147 to Đ238, the Standard tier from Đ238 to Đ340, the High tier from Đ340 to Đ502, and the Very High tier from Đ502 to Đ5,645. That top-end spread is especially important. It shows that some workforce segments, particularly in Helionis and Oryn, are not just marginally riskier than the safer population. They are in a different severity regime altogether.

The statutory-style benefit basis also matters. Benefits begin at a higher replacement rate in the first entitlement period and step down in the second, subject to the cap used in the model. This helps us match product design to claim duration. Roles exposed to longer absences, more complex recovery, or a harsher operating environment are not simply charged more because their salary is higher. They are charged more because the expected cost structure of the claim is heavier.

Table D.1.1 Workers' Compensation product ladder

Product form	Risk tiers	Charged premium band	Typical underwriting position
Workers Compensation ActiveCare Essentials	Very Low	Đ62 to Đ147	Lower-risk workforce pricing
Workers Compensation ActiveCare	Low to Standard	Đ147 to Đ340	Standard workforce pricing
Workers Compensation ActiveCare Plus	High to Very High	Đ340 to Đ5,645	Higher-risk workforce pricing

Table D.1.2 Workers' Compensation risk-tier schedule by system and workforce segment (full table)

Solar system	Risk tier	Product form	Target workforce	Charged premium band

Bayesia System	Very Low	ActiveCare Essentials	Technology Officer, Spacecraft Operator, Planetary Operations	62 to 147
Bayesia System	Low	ActiveCare	Drill Operator, Maintenance Staff, Safety Officer, Executive	147 to 238
Bayesia System	Standard	ActiveCare	Engineer, Administrator, Drill Operator (contract)	238 to 340
Bayesia System	High	ActiveCare Plus	Manager, Scientist	340 to 502
Helionis Cluster	Very Low	ActiveCare Essentials	Maintenance Staff, Spacecraft Operator, Planetary Operations	62 to 147
Helionis Cluster	Low	ActiveCare	Technology Officer, Safety Officer, Maintenance Staff	147 to 238
Helionis Cluster	Standard	ActiveCare	Administrator, Scientist, Technology Officer	238 to 340
Helionis Cluster	High	ActiveCare Plus	Administrator (contract), Drill Operator	340 to 502
Helionis Cluster	Very High	ActiveCare Plus	Executive, Manager, Engineer	502 to 5,645
Oryn Delta	Low	ActiveCare	Planetary Operations, Spacecraft Operator (contract)	147 to 238
Oryn Delta	Standard	ActiveCare	Spacecraft Operator (full time), Safety Officer	238 to 340
Oryn Delta	High	ActiveCare Plus	Administrator, Technology Officer, Maintenance Staff (contract)	340 to 502
Oryn Delta	Very High	ActiveCare Plus	Scientist, Manager, Drill Operator, Engineer, Maintenance Staff, Executive	502 to 5,645

Table D.1.3 Workers' Compensation risk-tier charged premium cut-offs

Risk tier	Charged premium lower bound	Charged premium upper bound
Very Low	62	147
Low	147	238
Standard	238	340
High	340	502
Very High	502	5,645

The full schedule shows why this segmentation is commercially useful. In Bayesia, lower-risk roles such as Technology Officers, Spacecraft Operators, and Planetary Operations sit in the Essentials end of the ladder. Managers and Scientists sit much higher. In Helionis, the upper-risk tail is strongest, with Executive, Manager, and Engineer roles occupying the Very High segment. In Oryn Delta, the upper-risk band applies to a broad set of roles because environmental and operational conditions amplify both injury and recovery risk. This means the line cannot be priced credibly as a single uniform payroll product.

D.2 Pricing and Capital Model

D.2.1. Feature Engineering & Variable Definition

The Workers' Compensation model begins with the core workforce, job-risk, and claims variables: occupation, employment_type, experience_yrs, accident_history_flag, psych_stress_index, hours_per_week, supervision_level, gravity_level, safety_training_index, protective_gear_quality, base_salary, exposure, claim_count, and claim_amount. These inputs are used to distinguish between the main drivers of claim frequency, claim duration, and benefit severity, rather than treating workers' compensation risk as a single uniform process.

To better capture these relationships, the model uses several engineered variables: salary_k, rookie_flag, high_gravity_flag, stress_gap, gear_deficit, workload_gravity, accident_risk_score, and annual_claim_rate. Each is intended to reflect a practical risk mechanism within the portfolio. For example, rookie_flag captures the higher injury risk often associated with less experienced workers, high_gravity_flag isolates roles with more physically demanding work, stress_gap reflects the imbalance between psychological strain and available training support, and gear_deficit captures reduced protection from lower-quality equipment. workload_gravity and accident_risk_score allow the model to reflect compounding effects, where longer hours, more hazardous work, prior incidents, and psychological strain interact to increase claims risk.

The pricing appendix then extends these model outputs into benefit-related variables such as avg_pred_claim_length_days, avg_pred_income_benefit, avg_total_cost_multiplier, avg_income_replacement_pct_of_annual_salary, avg_income_replacement_pct_during_absence, avg_pred_income_benefit_per_month_out, and benefit_period_profile. These outputs are important because they translate the model from pure claims incidence into benefit outcomes, allowing the analysis to assess wage replacement, duration strain, and recovery cost rather than focusing only on raw claim counts

D.2.2.Further Explanation

We price Workers' Compensation as a smaller but still profitable supporting line. In the short term, expected annual loss is 8.03m, projected claims cost is 8.39m, charged premium is 12.5m, and annual net revenue is 2.13m. In the long term, projected claims cost rises to 14.81m, charged premium rises to 22.0m, and annual net revenue rises to 3.8m. The line remains profitable, but on a scale that is much smaller than Equipment Failure.

The aggregate-loss results show a clear but tighter deterioration from Best to Moderate to Worst_1in100. In the short term, mean loss rises from 7.17m to 8.39m to 10.2m, P95 from 7.99m to 9.29m to 11.3m, P99 from 8.51m to 9.87m to 12.0m, and TVaR99 from 8.88m to 10.3m to 12.5m. In the long term, mean loss rises from 12.7m to 14.8m to 18.0m, P95 from 14.1m to 16.4m to 20.0m, P99 from 15.0m to 17.5m to 21.2m, and TVaR99 from 15.7m to 18.2m to 22.0m.

The stress results show that the line is driven by the interaction of injury frequency, time out of work, and benefit severity. A fatigue / overtime shock lifts mean loss by 8%. A recovery-duration shock lifts it by about 12%. A benefit-severity shock lifts it by about 15%. The combined workforce

shock lifts it by 25%. This is the numerical reason we do not rely on payroll alone. A role with a materially heavier duration profile should not be priced as though salary is the only driver.

The recovery-duration shock is particularly helpful for explaining the pricing logic. It raises short-term mean loss from 8.39m to 9.39m and long-term mean loss from 14.8m to 16.6m. It also lifts short-term P99 from 9.87m to 11.1m and long-term P99 from 17.5m to 19.6m. That is a meaningful change from a single deterioration in time out of work. Roles that are more exposed to long absences therefore need to be pushed higher in the product ladder.

Dependency is kept in place through the same common-shock framework used elsewhere in the portfolio. The effect is smaller than for Equipment Failure, but still real. Moving from the Moderate case to the Worst_1in100 case increases short-term mean loss by 21.5%, short-term P99 by 21.9%, and short-term TVaR99 by 21.6%. The long-term uplifts are 21.5%, 21.6%, and 21.3% respectively.

D.2.3. Methodology

Best uses 0.90x injury frequency, 0.95x severity, and no shared event. Moderate uses 1.00x frequency, 1.00x severity, and no shared event. Worst_1in100 uses 1.10x frequency, 1.10x severity, and a 1% shared-event overlay. Short-term mean loss moves from 7.17m to 8.39m to 10.2m. Long-term mean loss moves from 12.7m to 14.8m to 18.0m.

The stress tests also change specific variables. Fatigue / overtime shock increases hours_per_week and short-run incident pressure. Recovery duration shock increases predicted claim_length_days, so benefit duration rises. Benefit severity shock increases cost per claim through wage replacement, medical, and rehabilitation cost. Combined workforce shock applies all three.

Table D.2.1 Workers' Compensation pricing and capital summary

Horizon	Expected annual loss (m)	Projected claims cost (m)	Charged premium (m)	Annual net revenue (m)	Moderate mean loss (m)	Moderate P99 (m)	Worst-case TVaR99 (m)
Short term	8.03	8.39	12.5	2.13	8.39	9.87	12.5
Long term	8.03	14.81	22.0	3.8	14.8	17.5	22.0

Table D.2.2 Workers' Compensation scenario definitions

Scenario	Definition	Interpretation in model
Best	Smooth operations with attritional claims only	Lower-frequency, lower-severity workforce environment with no correlated catastrophe overlay
Moderate	Base workforce conditions	Central case used for pricing, capital, and stress comparison
Worst_1in100	Correlated catastrophe with shared event	Multi-system common-shock event that lifts injury incidence and severity simultaneously

Table D.2.3 Workers' Compensation aggregate loss results

Horizon	Scenario	Mean loss (m)	P95 (m)	P99 (m)	TVaR99 (m)
Short term	Best	7.17	7.99	8.51	8.88
Short term	Moderate	8.39	9.29	9.87	10.3
Short term	Worst_1in100	10.2	11.3	12.0	12.5
Long term	Best	12.7	14.1	15.0	15.7
Long term	Moderate	14.8	16.4	17.5	18.2
Long term	Worst_1in100	18.0	20.0	21.2	22.0

Table D.2.4 Workers' Compensation stress tests

Stress test	Horizon	Base mean loss (m)	Stressed mean loss (m)	Mean uplift	P95 (m)	P99 (m)	TVaR99 (m)
Fatigue / overtime shock	Short term	8.39	9.06	+8%	10.0	10.7	11.1
Recovery duration shock	Short term	8.39	9.39	+11.9%	10.4	11.1	11.5
Benefit severity shock	Short term	8.39	9.64	+14.9%	10.7	11.4	11.8
Combined workforce shock	Short term	8.39	10.5	+24.9%	11.6	12.3	12.8
Fatigue / overtime shock	Long term	14.8	16.0	+8%	17.7	18.9	19.6
Recovery duration shock	Long term	14.8	16.6	+12%	18.4	19.6	20.4
Benefit severity shock	Long term	14.8	17.0	+15%	18.9	20.1	20.9
Combined workforce shock	Long term	14.8	18.5	+25%	20.5	21.8	22.7

Table D.2.5 Workers' Compensation dependency test

Horizon	Moderate mean loss (m)	Worst_1in100 mean loss (m)	Mean uplift	Moderate P99 (m)	Worst_1in100 P99 (m)	P99 uplift	Moderate TVaR99 (m)	Worst_1in100 TVaR99 (m)	TVaR99 uplift
Short term	8.39	10.2	21.5%	9.87	12.0	21.9%	10.3	12.5	21.6%
Long term	14.8	18.0	21.5%	17.5	21.2	21.6%	18.2	22.0	21.3%

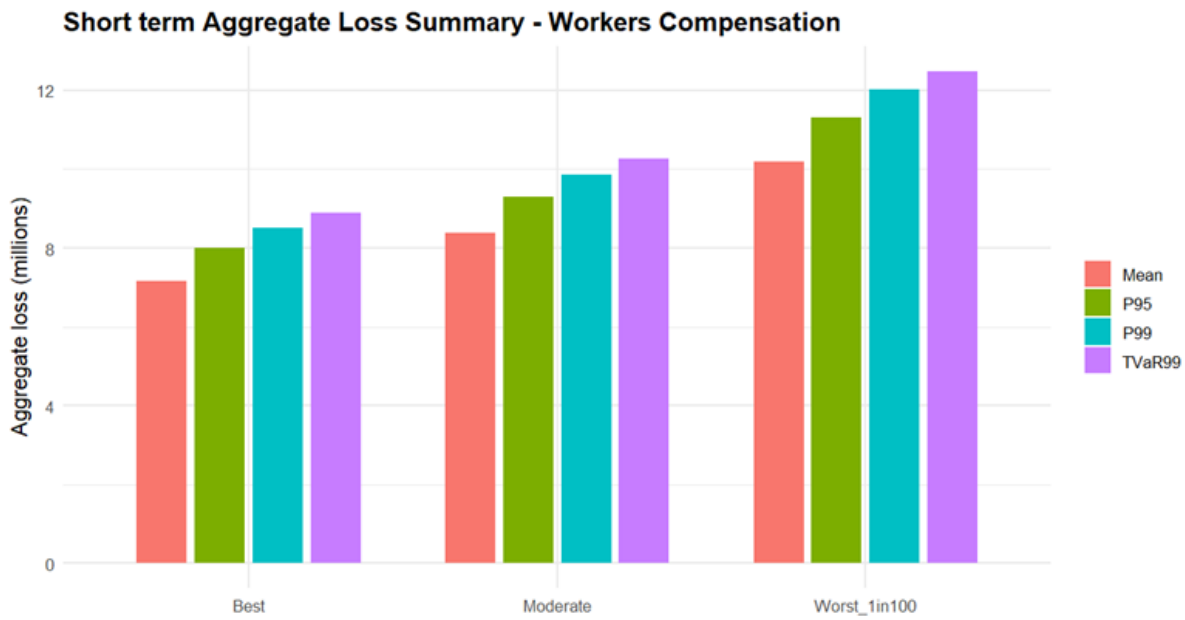


Figure D.2.1 Short-term Workers' Compensation aggregate loss summary.

Figure D.2.1 shows the same ordering as Equipment Failure but on a much tighter scale. The worst-case tail is visibly higher than the base case, but the absolute gap is far smaller than in the machinery line.

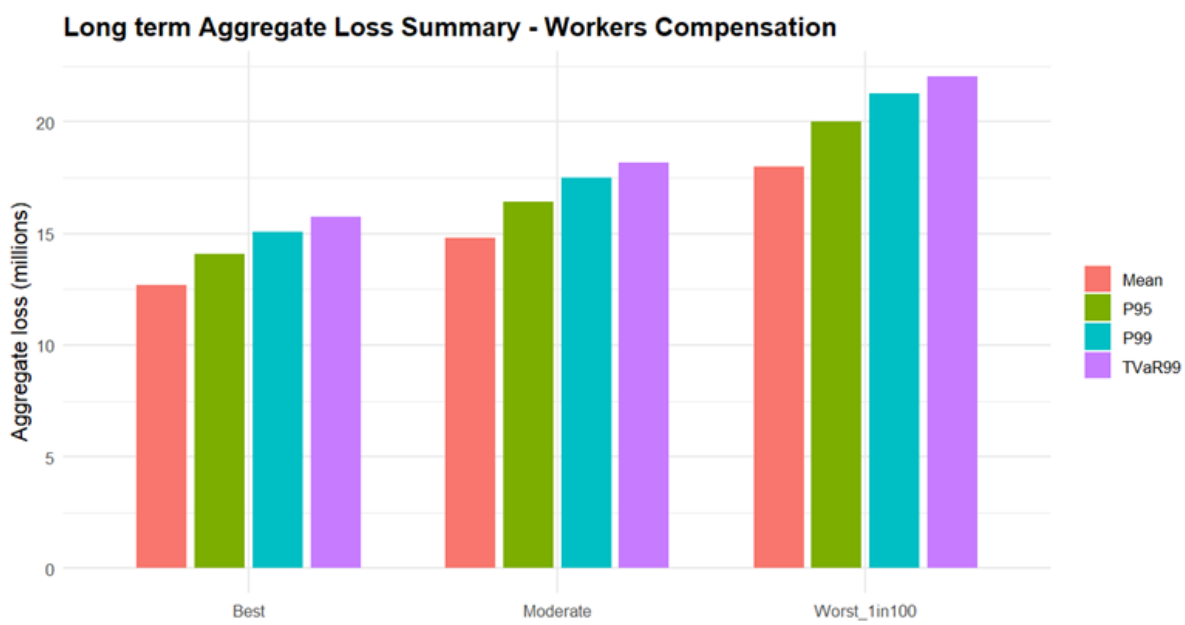


Figure D.2.2 Long-term Workers' Compensation aggregate loss summary.

Figure D.2.2 shows how long-term duration and severity accumulation widen the distribution. The long-term worst case remains manageable relative to Equipment Failure, but the uplift is still large enough to justify differentiated pricing and reserving.

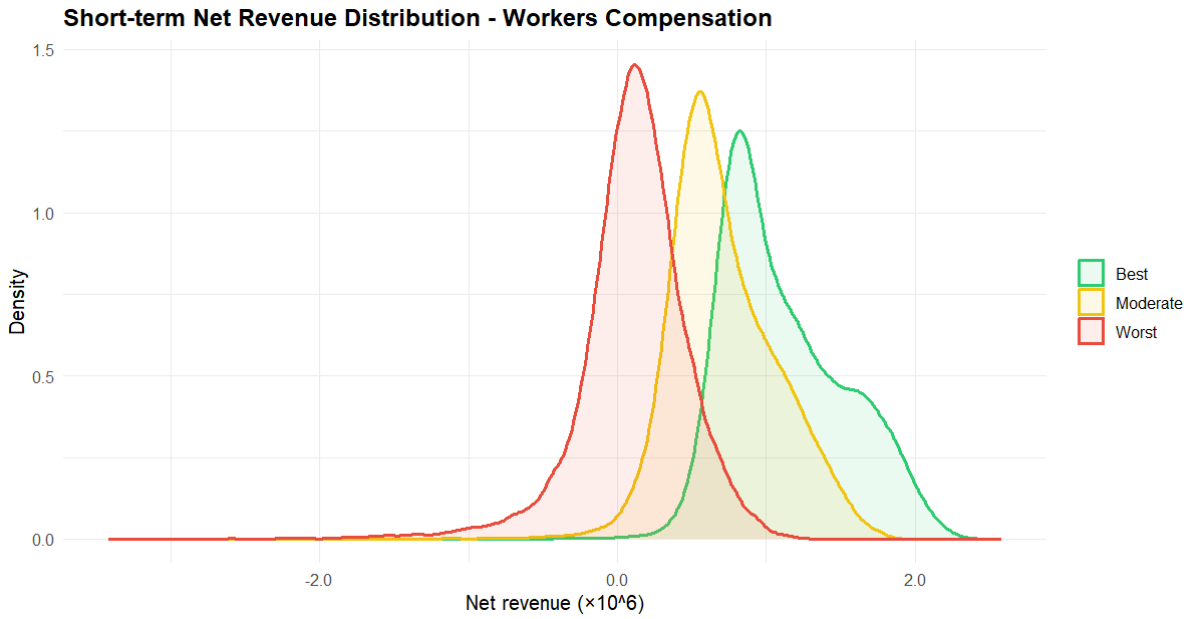


Figure D.2.3 Short-term Workers' Compensation net revenue distribution by scenario.

Figure D.2.3 shows that the line remains positive on average in all three scenarios, but the worst case pushes much more probability mass toward break-even or small losses. This is why Workers' Compensation is better viewed as a stable supporting line than as the main profit engine.

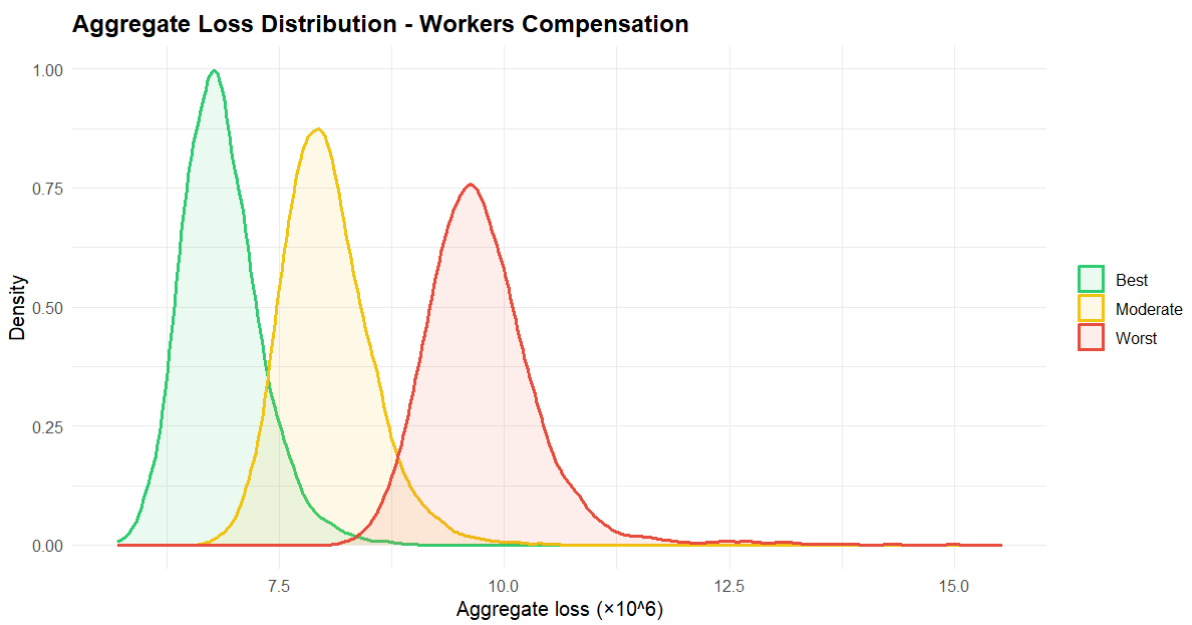


Figure D.2.4 Workers' Compensation aggregate loss density by scenario.

Figure D.2.4 provides the distributional view behind the bar summaries. The line clearly worsens under stress, but the density remains much more concentrated than in Equipment Failure.

D.3 Risk Assessment

D.3.1 Feature Engineering & Variable Definition

The Workers' Compensation model begins with the core workforce, job-risk, and claims variables: occupation, employment_type, experience_yrs, accident_history_flag, psych_stress_index, hours_per_week, supervision_level, gravity_level, safety_training_index, protective_gear_quality, base_salary, exposure, claim_count, and claim_amount. These inputs are used to distinguish between the main drivers of claim frequency, claim duration, and benefit severity, rather than treating workers' compensation risk as a single uniform process.

To better capture these relationships, the model uses several engineered variables: salary_k, rookie_flag, high_gravity_flag, stress_gap, gear_deficit, workload_gravity, accident_risk_score, and annual_claim_rate. Each is intended to reflect a practical risk mechanism within the portfolio. For example, rookie_flag captures the higher injury risk often associated with less experienced workers, high_gravity_flag isolates roles with more physically demanding work, stress_gap reflects the imbalance between psychological strain and available training support, and gear_deficit captures reduced protection from lower-quality equipment. workload_gravity and accident_risk_score allow the model to reflect compounding effects, where longer hours, more hazardous work, prior incidents, and psychological strain interact to increase claims risk.

The pricing appendix then extends these model outputs into benefit-related variables such as avg_pred_claim_length_days, avg_pred_income_benefit, avg_total_cost_multiplier, avg_income_replacement_pct_of_annual_salary, avg_income_replacement_pct_during_absence, avg_pred_income_benefit_per_month_out, and benefit_period_profile. These outputs are important because they translate the model from pure claims incidence into benefit outcomes, allowing the analysis to assess wage replacement, duration strain, and recovery cost rather than focusing only on raw claim counts.

D.3.2 Further Explanation

Workers' Compensation risk differs by solar system, but on a much smaller scale than Equipment Failure. Helionis Cluster has the largest short-term expected annual loss at 3.66m and is driven by extraction injury, transport incident, and debris-linked trauma. Oryn Delta follows at 2.53m and is defined by low-visibility incident, fatigue, and delayed rescue. Bayesia System is smaller at 1.85m, but remains exposed to gravity strain, radiation exposure, and heat stress.

Correlated risk still matters. A common-shock event that disrupts multiple systems at once can raise injury incidence, stretch rehabilitation and rescue, and increase recovery duration across more than one solar system. We therefore keep the common-shock catastrophe framework for Workers' Compensation as well, even though its capital effect is weaker than for Equipment Failure.

Scenario testing uses the same three-case structure. The Best case is smooth operations with attritional claims only, the Moderate case is the central workforce environment, and the Worst_1in100 case is a correlated catastrophe with a shared-event overlay. Under those definitions, short-term mean aggregate loss rises from 7.17m to 8.39m to 10.2m, while long-term mean aggregate loss rises from 12.7m to 14.8m to 18.0m. The shift is smaller than in the machinery line, but still large enough to change pricing and reserve posture.

The threat ranking shows that Workers' Compensation is driven by interacting workforce conditions rather than a single dominant catastrophe driver. Combined workforce shock ranks first, followed by benefit severity, recovery duration, and fatigue / overtime. The recovery-duration shock is especially important because it directly links pricing to time out of work. If duration worsens, the line becomes more expensive even without a dramatic change in claim count.

The system narrative follows from that ranking. Helionis is the largest absolute Workers' Compensation market, Oryn Delta remains operationally fragile because fatigue and delayed rescue can amplify claim duration, and Bayesia remains smaller but clearly affected by the physical operating environment. This supports differentiated role and system pricing rather than a uniform payroll product.

D.3.3 Methodology

Scenario testing uses the same three states. Best uses 0.90x frequency, 0.95x severity, and no shared event. Moderate uses 1.00x frequency, 1.00x severity, and no shared event. Worst_1in100 uses 1.10x frequency, 1.10x severity, and a 1% shared-event overlay with system catastrophe multipliers. Short-term mean loss moves from 7.17m to 8.39m to 10.2m. Long-term mean loss moves from 12.7m to 14.8m to 18.0m.

The threat table also maps directly to the variables. Fatigue / overtime shock increases hours_per_week and incident pressure. Recovery duration shock increases predicted claim_length_days and slows return-to-work. Benefit severity shock increases cost per claim through wage replacement, medical, and rehabilitation cost. Combined workforce shock applies all three. Helionis is the main exposure base. Oryn Delta is the most duration-sensitive. Bayesia remains exposed to gravity, heat, and radiation-related strain.

Table D.3.1 Workers' Compensation top threats

Rank	Threat	Mean loss uplift vs Moderate
1	Combined workforce shock	+25%
2	Benefit severity shock	+15%
3	Recovery duration shock	+12%
4	Fatigue / overtime shock	+8%

Table D.3.2 Workers' Compensation solar-system risk profile

Solar system	Primary threat	Short-term expected annual loss (m)

Helionis Cluster	Extraction injury / transport incident / debris-linked trauma	3.66
Bayesia System	Gravity strain / radiation exposure / heat stress	1.85
Oryn Delta	Low-visibility incident / fatigue / delayed rescue	2.53

Appendix E – Cargo Loss

E.1 Product Design

Product premiums tabulated below:

Product Name	***Route Risk***	***Premium (Nearest Whole Number)***
Low Cargo Cover	1	Đ1 152 845
Core Transit Cover	2	Đ1 585 459
Shipment Essentials	3	Đ2 148 708
High Transit Cover	4	Đ2 569 864
Extreme Cargo Cover	5	Đ3 220 944

E.2 Pricing and Capital

E.2.1 Methodology

The aggregate loss distributions must be contextualised in terms of the frequency and severity models used to develop it. The claim frequency model was determined by comparing Poisson and Negative Binomial generalised linear models (GLMs). These two models were selected as they are historically used in insurance, and for a brand-new product, will minimise model risk until we have more information and can develop the product further. Based on model performance, the Poisson model had the lower score in terms of RMSE and MAE metrics, so the more accurate model was chosen. Further to reduce potential overfitting, stepwise AIC selection was applied to the Poisson model to further improve accuracy. For similar reasons, claim severity compared a Gamma and log-linear model, eventually determining that the AIC stepwise nested Gamma model to be the most favourable.

In each Monte Carlo simulation, claim counts generated from the Poisson distribution multiply with the severity of the gamma model to find the expected loss for each policy. The total portfolio loss is then calculated by summing the expected loss of each policy. Repeating this process 500 times produces the short-term distribution above. For long-term losses, the Monte Carlo simulation is extended over a 10-year period and adjusted for inflation, using historical data. In other words, it is assumed claim severity and premiums grow with the rate of inflation. Like before the total portfolio losses are found, except now for each year and then discounted to be in terms of present value, using historical 1-year spot rates.

Frequency Model Comparison

Model Complexity Comparison	
Model	***Statistic***
Full Poisson	AIC: 97236.99 BIC: 97508.41
Full Negative Binomial	AIC: 96707.99 BIC: 96988.76

We note above that there is a very small difference in terms of simplicity between the models.

Model Accuracy Comparison	
Model	***Statistic***
Full Poisson	RMSE: 0.5492929 MAE: 0.3478251
Full Negative Binomial	RMSE: 0.5514125 MAE: 0.3497327
AIC Nested Poisson	RMSE: 0.5492325 MAE: 0.3478593

```
> summary(PoissonModelAIC)

Call:
glm(formula = claim_count ~ cargo_value + weight + route_risk +
    pilot_experience + vessel_age + container_type + exposure +
    container_utilisation + Fragility + adjusted_risk + vessel_stress,
    family = poisson(link = "log"), data = TrainData)

Coefficients:
                                    Estimate Std. Error z value Pr(>|z|)
(Intercept)                        3.759e-01  4.860e-02   7.734 1.04e-14 ***
cargo_value                       -2.405e-10  1.324e-10  -1.816 0.069319 .
weight                             4.244e-07  1.727e-07   2.458 0.013984 *
route_risk2                       -4.084e-02  2.987e-02  -1.367 0.171515
route_risk3                       -1.271e-01  3.272e-02  -3.884 0.000103 ***
route_risk4                       -2.367e-01  4.217e-02  -5.613 1.99e-08 ***
route_risk5                       -3.610e-01  5.452e-02  -6.622 3.54e-11 ***
pilot_experience                  -6.078e-03  1.382e-03  -4.397 1.10e-05 ***
vessel_age                        -2.554e-03  1.013e-03  -2.521 0.011697 *
container_typeDockArc Freight Case -2.948e-02  2.497e-02  -1.181 0.237774
container_typeHardSeal Transit Crate  2.176e-02  2.810e-02   0.774 0.438876
container_typeLongHaul Vault Canister -2.955e-02  2.962e-02  -0.998 0.318357
container_typeQuantumCrate Module  5.616e-03  3.060e-02   0.184 0.854385
exposure                          -5.168e+00  1.105e-01 -46.748  < 2e-16 ***
container_utilisation             -1.108e-02  7.414e-03  -1.494 0.135157
Fragility                          1.003e-06  5.285e-07   1.897 0.057807 .
adjusted_risk                      4.612e-01  3.321e-02  13.890  < 2e-16 ***
vessel_stress                      8.200e-03  2.904e-03   2.824 0.004748 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 75445  on 85739  degrees of freedom
Residual deviance: 62533  on 85722  degrees of freedom
AIC: 97225

Number of Fisher Scoring iterations: 6
```

Comparing the above tabulated results clearly indicates that the Poisson model is more accurate than the Negative Binomial function. On the left we observe how performing stepwise selection with the AIC significantly simplified the model. This simplification as well as preventing overfitting, also assists with feature selection and the identification of key risks to CL.

Severity Model Comparison

Model Complexity Comparison	
Model	***Statistic***
Full Gamma	AIC: 625565.6 BIC: 625804.7
Full Log Linear	AIC: 52058.9 BIC: 52298.05

Model Accuracy Comparison	
Model	***Statistic***

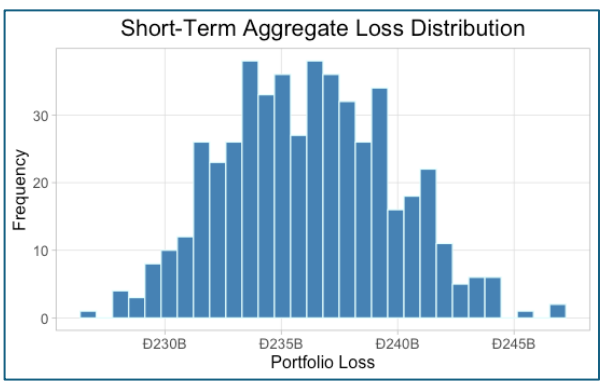
Full Gamma	RMSE: 14918676 MAE: 4647978
Full Log Linear	RMSE: 22600803 MAE: 7621684
AIC Gamma	RMSE: 14913448 MAE: 4648146

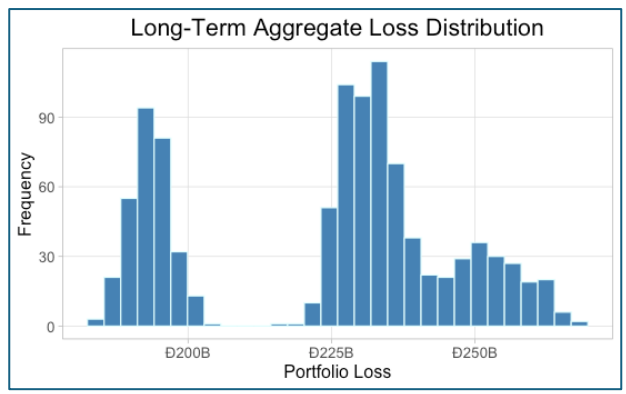
Here we notice that there is a fairly substantial difference between the complexity of a gamma and log-linear model. However, in turn there is also a significant difference between the accuracy between the distributions. In this case, we will always choose the more accurate model, as we value accuracy for this task over interpretability.

```
> summary(GammaModelAIC)

Call:
glm(formula = claim_amount ~ cargo_type + cargo_value + weight +
    route_risk + solar_radiation + debris_density + exposure,
    family = Gamma(link = "log"), data = TrainData)

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            1.165e+01  5.551e-02 209.838  < 2e-16 ***
cargo_typegold         4.116e+00  9.209e-02  44.690  < 2e-16 ***
cargo_typelithium      4.445e-01  2.860e-02  15.538  < 2e-16 ***
cargo_typeplatinum     3.579e+00  5.370e-02  66.651  < 2e-16 ***
cargo_typerare earths -3.267e-02  4.197e-02  -0.779   0.4363
cargo_typesupplies    -1.779e+00  3.390e-02 -52.464  < 2e-16 ***
cargo_typetitanium    -2.037e+00  2.822e-02 -72.192  < 2e-16 ***
cargo_value            2.182e-09  1.532e-10  14.250  < 2e-16 ***
weight                 7.432e-06  2.894e-07  25.677  < 2e-16 ***
route_risk2            2.233e-01  2.885e-02   7.740 1.04e-14 ***
route_risk3            4.242e-01  2.670e-02  15.888  < 2e-16 ***
route_risk4            5.438e-01  2.909e-02  18.690  < 2e-16 ***
route_risk5            7.432e-01  3.329e-02  22.327  < 2e-16 ***
solar_radiation        4.783e-01  6.401e-02   7.472 8.21e-14 ***
debris_density         2.862e-01  6.784e-02   4.219 2.47e-05 ***
exposure               6.387e-02  2.839e-02   2.250   0.0245 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 1.055671)

    Null deviance: 108442  on 21408  degrees of freedom
Residual deviance:  14419  on 21393  degrees of freedom
AIC: 625548

Number of Fisher Scoring iterations: 5
```

Like for the frequency model, we perform AIC stepwise selection to prevent overfitting and assist with feature selection. Interestingly, we observe how the frequency and severity models utilise different variables for their predictions.

E.2.2 Aggregate Loss Graphs





E.2.3 Stress Testing

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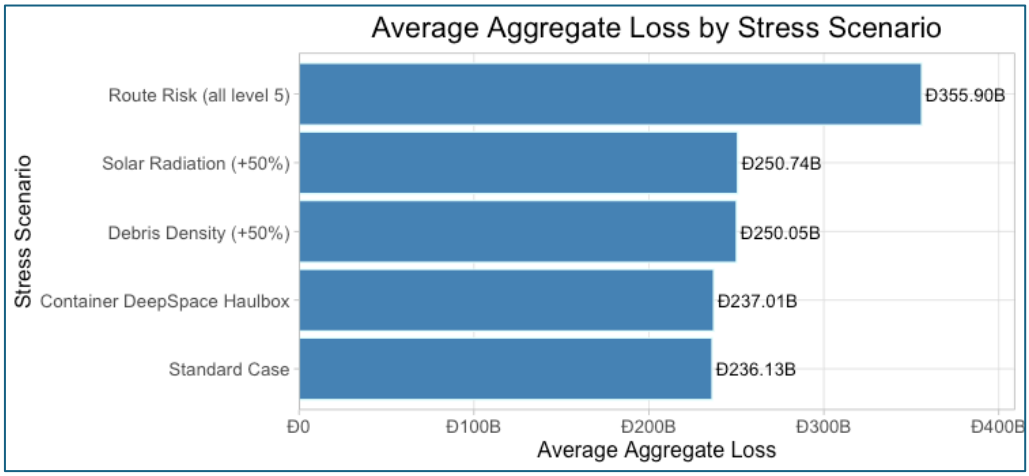


Several adverse scenarios were created assess how aggregate losses are affected by key risk drivers. A solar flare occurrence was modelled by increasing 'Solar Radiation' by 50%, reflecting the possibility that an abnormal radiation spike disrupts navigation. Unexpected orbital resonances may "shuffle" rubble in space making for a more hazardous transit and is depicted by a 50% increase in 'Debris Density'. Altering 'Solar Radiation' and 'Debris Density' similarly impacts claims, modestly increasing them to Đ250B and Đ249. Further, a supply chain disruption could render all the stronger containers unavailable, forcing the overloading of smaller container types, depicted by changing the 'Container Type' variable to be DeepSpace Haulbox for every policy. Interestingly, alone this change does not significantly impact aggregate losses for the portfolio. Finally, an intergalactic incident such as war, might leave major routes closed, forcing all shipments into the highest route-risk category. This variable would cause the biggest disturbance upon claims, clearly increasing expected aggregated losses by the greatest amount to Đ355B. These results have important implications for underwriting, pricing and capital management, as it showcases how the product can become volatile in extreme catastrophic events, thus highlighting the importance of reserving capital to act as a buffer for uncertainty.

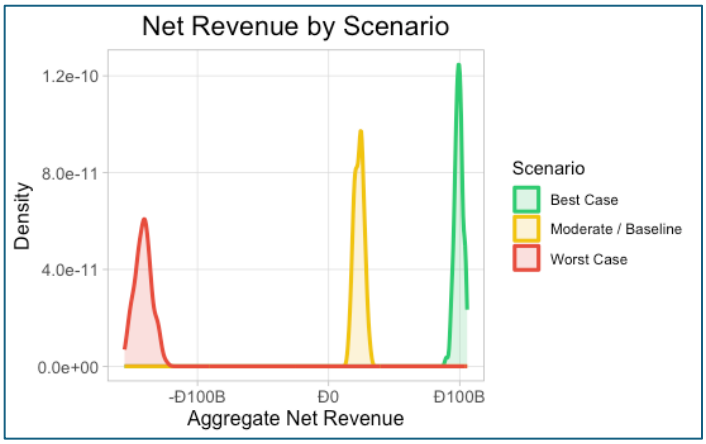
E.3 Risk Assessment

E.3.1 Identification of Cargo Risks

For Cargo Cover, the following risks will differ across solar systems. The Zeta system experiences a higher solar flare risk due to its erratic central star. Space Debris Risk is the chance that space junk acts sporadically and damages cargo enroute. It is prevalent among all three operational solar systems, but history suggests the Hilionis cluster experiences it the worst, followed by the Zeta and Epsilon systems. The subsequent risks are all fairly likely to occur in all three solar systems and are factors which relate less to the environment and more to operational flaws. Route risk refers to the fact that cargo's journey may unexpectedly become riskier, causing a mismatch of product and insured. Vessel stress occurs when a container is overloaded beyond its maximum capacity and refers to the probability of any damage caused by this. Finally, an inexperienced pilot may be inadequately prepared to handle unexpected troubles during transit and may induce losses where a more experienced captain would not have.

E.3.2 Scenario Testing

In CL testing, a moderate scenario corresponds to regular business operations working under the standard assumptions used in earlier simulations and pricing structures. The worst-case scenario is identified to be all the previously stress tested events occurring simultaneously. Meanwhile, a best-case scenario operates under the assumption that business operations will be smooth with only attritional claims. To reflect this scenario, the product will be modelled against a reduced claim frequency and severity by 20% and 15% respectively.



E.3.3 Capital Requirements

Reserved Type	***Amount (Đ)***
Standard Business Operations Reserves	7 247 947 158
Extreme Events Reserves	154 371 640 524

The results above were found at the 99% VaR level after considering aggregate loss for the moderate and worst-case scenarios. Importantly, under regular business operations, the revenue from premiums is more than enough to cover aggregate loss. In this way the 'reserves' are a figure of how much revenue needs to be retained each year to cover loss. In contrast, the reserves for extreme events are the capital requirements for the company to survive times of financial hardship. The reserves are significantly higher for extreme events as the reserves also buffer against the net loss the company will make in that catastrophic year.

E.4 Assumptions

The following assumptions are for factors which are assumed to be over their numerical ranges. For 'Weight', policies over the range were assumed to have been written in grams instead of kg, so they were divided by 1000. 'Distance' was assumed to have been calculated in centimetres instead of meters, so it was divided by 100. 'Transit Duration' was assumed to have been written in days instead of months. 'Pilot Experience' was assumed to have been written months instead of years. 'Vessel Age' was assumed to be written in weeks instead of days. 'Solar Radiation' and 'Debris Density' were assumed to have been incorrectly indexed, written from a scale of 1 to 10 instead of 0 to 1. For 'Route Risk' and 'Exposure' given a small number of policies were not correctly formatted in the range and creating an assumption to undo them would make them messy, so they were simply removed.

One crucial assumption previously stated is that the CL product can be applied to all shipments regardless of solar system. These assumptions needed to be made due to the data set not containing a solar system identifier, requiring us to consider other factors, such as 'Route Risk', for determining product segmentation. Building on this, it is also assumed there exists a strong dependency between solar systems, such that they will experience environmental and operational shocks simultaneously. This assumption is inferred since the product cannot be segmented according to the solar system.

Additionally, the Poisson and Gamma assumptions implicitly implied by model selection also have a large impact on pricing and capital reserves. These assumptions were taken on as these models proved to be the strongest for modelling frequency and severity respectively.

In long-term modelling, as established, we assume that premiums and claim severity both grow equally with inflation. Further we assume that the historic inflation and 1-year spot rate data is a reasonable proxy for future discounting. This assumption was made so that the model can be tested under realistic circumstances, and due to the data limitations, there was no other alternative.

In scenario testing, the assumptions made for the 'best' and 'worst' cases are relatively reasonable, as to explore how the product behaves in these potential disastrous events, extreme assumptions must be made.

Appendix F – Generative AI Usage

Artificial intelligence was utilised in the completion of this task, especially for troubleshooting coding errors. AI was able to provide reasoning for why errors were happening, how to fix them, and how to prevent them in the future. This allowed for a more comprehensive understanding of the task at hand and dealing with data on the whole.

Let it be known that AI was not utilised for analysing outputs, forming conclusions, designing products, and the physical writing of this report.